Physics

PROBLEMS

1. A long, thin, pliable carpet is laid on the floor. One end of the carpet is bent back and then pulled backwards with constant unit velocity, just above the part of the carpet which is still at rest on the floor.

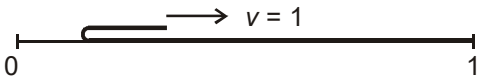


Fig. 1

Find the speed of the centre of mass of the moving part. What is the minimum force needed to pull the moving part, if the carpet has unit length and unit mass?

2. A man of height h_0 = 2 m is bungee jumping from a platform situated a height h = 25 m above a lake. One end of an elastic rope is attached to his foot and the other end is fixed to the platform. He starts falling from rest in a vertical position.

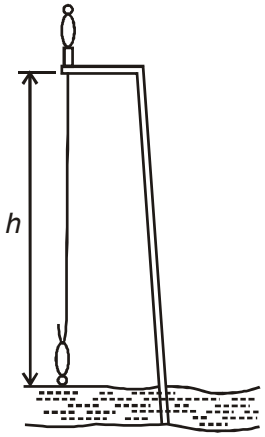


Fig. 2

The length and elastic properties of the rope are chosen so that his speed will have been reduced to zero at the instant when his head reaches the surface of the water. Ultimately the jumper is hanging from the rope, with his head 8 m above the water.

(i) Find the unstretched length of the rope.

(ii) Find the maximum speed and acceleration achieved during the jump.

3. A sphere, made of two non-identical homogeneous hemispheres stuck together, is placed on a plane inclined at an angle of 30° to the horizontal. Can the sphere remain in equilibrium on the inclined plane?

4. If the mass of each of the members of a binary star were the same as that of the Sun, and their distance apart were equal to the Sun-Earth distance, what would be their period of revolution?

5. Two (non-physics) students, *A* and *B*, living in neighbouring college rooms, decided to economise by connecting their ceiling lights in series. They agreed that each would install a 100 W bulb in their own rooms and that they would pay equal shares of the electricity bill. However, both decided to try to get better lighting at the other's expense, *A* installed a 200 W bulb and *B* installed a 50 W bulb. Which student subsequently failed the end-of-term examinations?

6. A circular hole of radius *r* at the bottom of an initially full water container is sealed by a ball of mass *m* and radius *R* (> *r*). The depth of the water is now slowly reduced and when it reaches a certain value, h_0, the ball rises out of the hole. Find h_0.

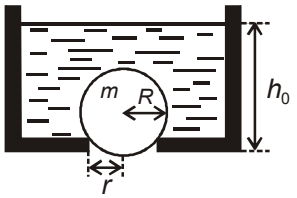


Fig. 4

7. A tree-trunk of diameter 20 cm lies in a horizontal field. A lazy grasshopper wants to jump over the trunk. Find the minimum take-off speed of the grasshopper that will suffice. (Air resistance is negligible.)

8. A plate, bent at right angles along its centre line, is placed onto a horizontal fixed cylinder of radius *R* as shown in the figure.

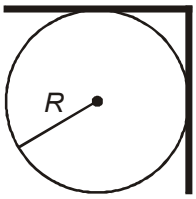


Fig. 5

How large does the coefficient of static friction between the cylinder and the plate need to be if the plate is not to slip off the cylinder?

9. An executive toy consists of three suspended steel balls of masses *M*, μ and *m* arranged in that order with their centres in a horizontal line. The ball of mass *M* is drawn aside in their common plane until its centre has been raised by *h* and is then released. If $M \neq m$ and all collisions are elastic, how must μ be chosen so that the ball of mass *m* rises to the greatest possible height? What is this height? (Neglect multiple collisions.)

10. After a tap above and empty rectangular basin has been opened, the basin fills with water in a time T_1. After the tap has been closed, opening a plug-hole at the bottom of the basin empties it in a time T_2. What happens if both the tap and the plug-hole are open? What ratio of $\frac{T_1}{T_2}$ can cause the basin to overflow? As a specific case, let T_1 = 3 minutes and T_2 = 2 minutes.

11. A glass prism whose cross-section is an isosceles triangle stands with its (horizontal) base in water, the angles that its two equal sides make with the base are each θ.

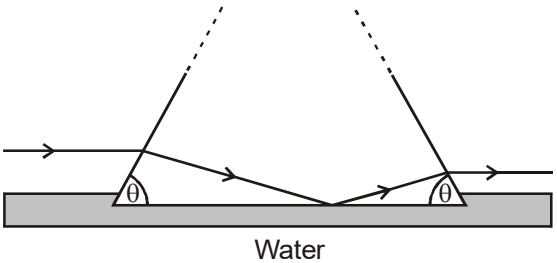


Fig. 6

An incident ray of light, above and parallel to the water surface and perpendicular to the prism's axis, is internally reflected at the glass-water interface and subsequently re-emerges into the air. Taking the refractive indices of glass and water to be $\frac{3}{2}$ and $\frac{4}{3}$ respectively, show that θ must be at least 25.9°.

12. A simple pendulum and a homogeneous rod pivoted at its end are released from horizontal positions. What is the ratio of their periods of swing if their lengths are identical?



Fig. 7

13. Water, which wets glass, is stuck between two parallel glass plates. The distance between the plates is d and the diameter of the trapped water 'disc' is $D >> d$.

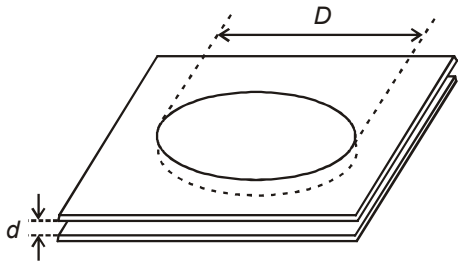


Fig. 9

What is the force acting between the plates?

14. A spider has fastened one end of a 'super-elastic' silk thread of length 1 m to a vertical wall. A small caterpillar is sitting somewhere on the thread.

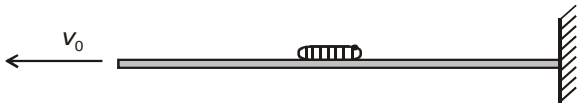


Fig. 10

The hungry spider, whilst not moving from its original position, starts pulling in the other end of the thread with uniform speed, v_0 = 1 cm s^{-1}. Meanwhile, the caterpillar starts fleeing towards the wall with a uniform speed of 1 mm s^{-1} with respect to the moving thread. Will the caterpillar reach the wall?

15. Find the angle to which a pair of compasses should be opened in order to have the pivot as elevated as possible when the compasses are suspended from a string attached to one of the points, as shown in the figure. Assume that the lengths of the compass arms are equal.

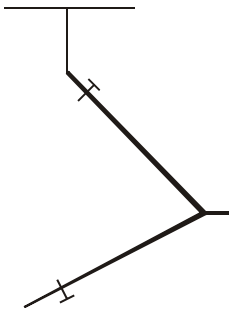


Fig. 11

16. A small bob joins two light unstretched, identical springs, anchored at their far ends and arranged along a straight line, as shown in the figure.

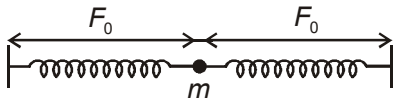


Fig. 15

The bob is displaced in a direction perpendicular to the line of the springs by 1 cm and then released. The period of the ensuing vibration of the bob is 2 s. Find the period of the vibration if the bob were displaced by 2 cm before release. The unstretched length of the springs is $\ell_0 >> 1$ cm, and gravity is be ignored.

17. One end of a light, weak spring, of unstretched length *L* and force constant *k*, is fixed to a pivot, and a body of mass *m* is attached to its other end. The spring is released from an unstretched, horizontal position, as in the figure.

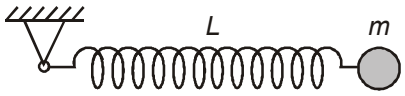


Fig. 16

What is the length of the spring when it reaches a vertical position? (Describing a spring as weak implies that *mg* >> *kL*, and that the tension in the spring is directly proportional to its extension at all times.)

18. A glass partially filled with water is fastened to a wedge that slides, without friction, down a large plane inclined at an angle α as shown in the figure. The mass of the inclined plane is *M*, the combined mass of the wedge, the glass and the water is *m*.

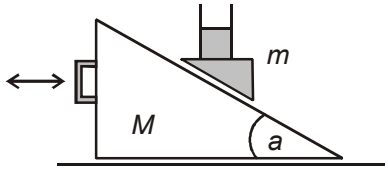


Fig. 17

If there were no motion the water surface would be horizontal. What angle will it ultimately make with the inclined plane if

(i) The inclined plane is fixed

(ii) The inclined plane can move freely in the horizontal direction?

Examine also the case in which *m* >> *M*. What happens if the handle of the inclined plane is shaken in a periodic manner, but one that is such that it does not cause the wedge to rise off the plane?

19. Point-masses of mass *m* are at rest at the corners of a regular *n*-gon, as illustrated in the figure for *n* = 6.

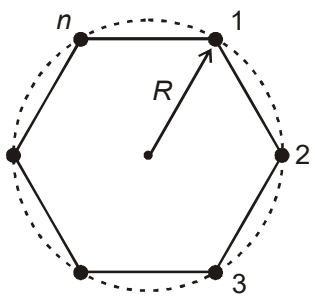


Fig. 18

How does the system move if only gravitation acts between the bodies? How much time elapses before the

bodies collide if n = 2, 3 and 10? Examine the limiting case when $n >> 1$ and $m = M_0/n$, where M_0 is a given total mass.

20. A long slipway, inclined at an angle α to the horizontal, is fitted with many identical rollers, consecutive ones being a distance d apart. The rollers have horizontal axles and consist of rubber-covered solid steel cylinders each of mass m and radius r. Planks of mass M, and length much greater than d, are released at the top of the slipway.

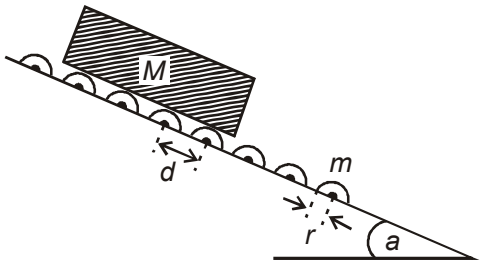


Fig. 19

Find the terminal speed v_{max} of the planks. Ignore air resistance and friction at the pivots of the rollers.

21. In a physics stunt, two balls of equal density, and radii r and $R = 2r$, are placed with the centre of the larger one at the middle of a cart of mass M = 6 kg and length L = 2 m. The mass of the smaller ball is m = 1 kg. The balls are made to roll, without slipping, in such a way that the larger ball rests on the cart, and a straight line connecting their centres remains at a constant angle $\phi = 60°$ to the horizontal. The cart is pulled by a horizontal force in the direction shown in the figure.

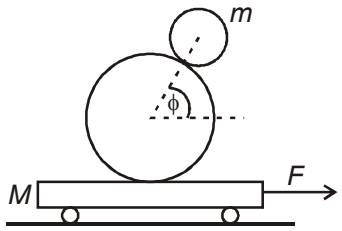


Fig. 20

(i) Find the magnitude of the force F.

(ii) How much time elapses before the balls fall off the cart?

22. A light, inelastic thread is stretched round one-half of the circumference of a fixed cylinder as shown in the figure.

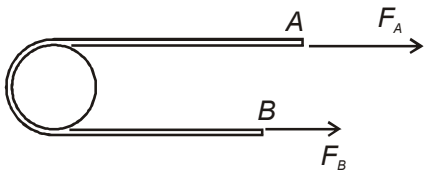


Fig. 21

As a result of friction, the thread does not slip on the cylinder when the magnitudes of the forces acting on its ends fulfil the inequality

$$\frac{1}{2}F_A \le F_B \le 2F_A.$$

Determine the coefficient of friction between the thread and the cylinder.

23. A table of height 1 m has a hole in the middle of its surface. A thin, golden chain necklace, of length 1 m, is placed loosely coiled close to the hole, as shown in the figure.

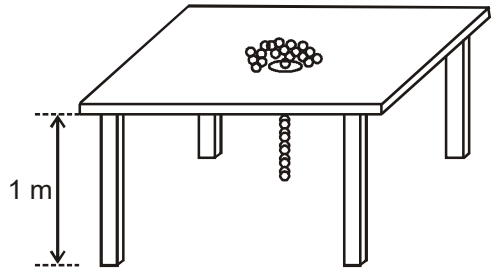


Fig. 22

One end of the chain is pulled a little way through the hole and then released. Friction is negligible and as a result, the chain runs smoothly through the hole with increasing speed. After what times will the two ends of the chain reach the floor?

24. A long, heavy, flexible rope with mass ρ per unit length is stretched by a constant force *F*. A sudden movement causes a circular loop to form at one end of the rope. In a manner similar to that in which transverse waves propagate, the loop runs (rolls) along the rope with speed *c* as shown in the figure.

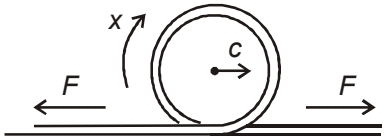


Fig. 24

(i) Calculate the speed *c* of the loop.

(ii) Determine the energy, momentum and angular momentum carried by a loop of angular frequency ω. What is the relationship between these quantities?

25. Sand falls vertically at a rate of 50 kg s^{-1} onto a horizontal conveyor belt moving at a speed of 1 ms^{-1}, as shown in the figure.

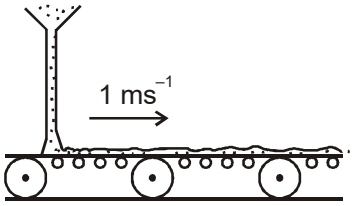


Fig. 25

What is the minimum power output of the engine which drives the belt? How is the work done by the engine accounted for?

26. A metal sphere, of radius *R* and cut in two along a plane whose minimum distance from the sphere's centre is *h*, is uniformly charged by a total electric charge *Q*. What force is necessary to hold the two parts of the sphere together?

27. Electrical charges are evenly distributed along a long, thin insulating rod *AB*.

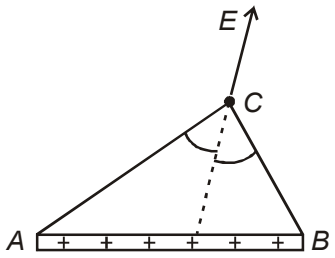


Fig. 27

Show that at an arbitrary point *C* (see figure), the electric field due to the rod points in the direction of the bisector of angle *ACB*.

28. The distance between the plates of a plane capacitor is *d* and the area of each plate is *A*. As shown in the figure, both plates of the capacitor are earthed and a small body carrying charge *Q* is placed between them, at a distance *x* from one plate.

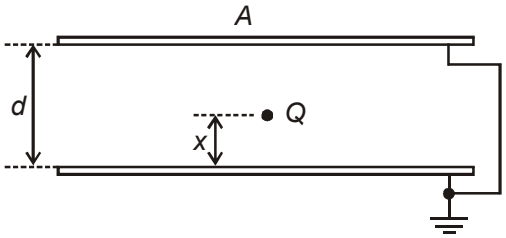


Fig. 28

What charge will accumulate on each plate?

29. The refractive index of the medium within a certain region, $x > 0$, $y > 0$, changes with *y*. A thin light ray travelling in the *x*-direction strikes the medium at right angles and moves through the medium along a circular arc.

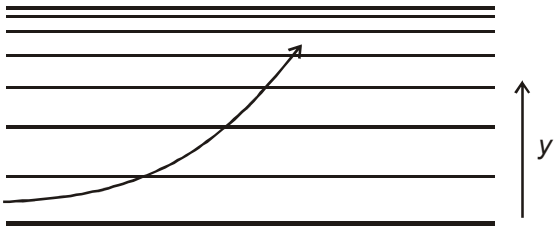


Fig. 29

How does the refractive index depend on *y*? What is the maximum possible angular size of the arc?

30. A closed body, whose surface *F* is made of metal foil, has an electrical capacitance *C* with respect to an 'infinitely distant' point. The foil is now dented in such a way that the new surface *F** is entirely inside or on the original surface, as shown in the figure.

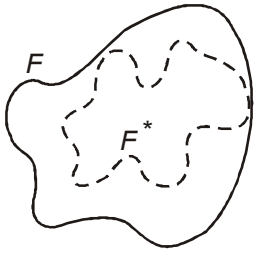


Fig. 30

Prove that the capacitance of the deformed body is less than *C*.

31. A grid in the shape of a regular polyhedron (tetrahedron, cube, dodecahedron, etc.) is made up of identical, say 1 Ω, resistors. What is the equivalent resistance between two neighbouring grid points?

32. One end of a horizontal track of gauge ℓ and negligible resistance, is connected to a capacitor of capacitance C charged to voltage V_0. The inductance of the assembly is negligible. The system is placed in a homogeneous, vertical magnetic field of induction B, as shown in the figure.

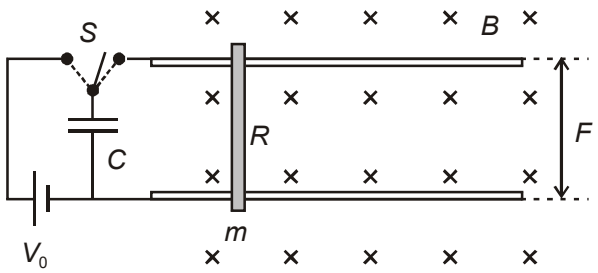


Fig. 32

A frictionless conducting rod of mass m and resistance R is placed perpendicularly onto the track. The polarity of the capacitor is such that the rod is repelled from the capacitor when the switch is turned over.

(i) What is the maximum velocity of the rod?

(ii) Under what conditions is the efficiency of this 'electromagnetic gun' maximal?

33. The circuit shown in the figure, consisting of three identical lamps and two coils – is connected to a direct current source. The ohmic resistance of the coils is negligible.

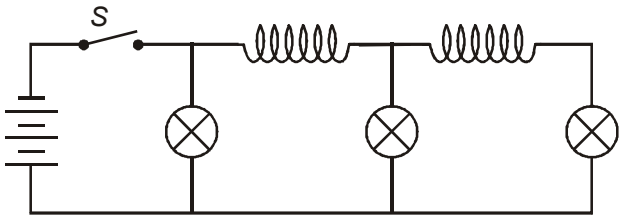


Fig. 34

After some time, switch S is opened. What are the relative brightnesses of the three lamps immediately afterwards?

34. A solid metal cylinder rotates with angular velocity ω about its axis of symmetry. The cylinder is in a homogeneous magnetic field B parallel to its axis. What is the resultant charge distribution inside the cylinder? Is there an angular velocity for which the charge density is everywhere zero?

35. A uniform thin wire of length $2\pi a$ and resistance r has its ends joined to form a circle. A small voltmeter of resistance R is connected by tight leads of negligible resistance to two points on the circumference of the circle at angular separation θ, as shown in the figures.

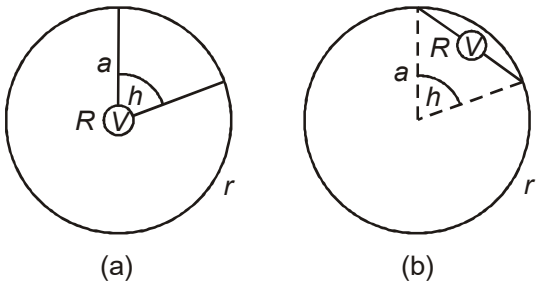


Fig. 35

A uniform magnetic flux density perpendicular to the plane of the circle is changing at a rate $\dot{B}$. What will the reading of the voltmeter be if the voltmeter is positioned :

(a) At the centre of the circle, and

(b) On the chord joining the two points of attachment?

36. A long solenoid contains another coaxial solenoid (whose radius R is half of its own). Their coils have the same number of turns per unit length and initially both carry no current. At the same instant currents start increasing linearly with time in both solenoids. At any moment the current flowing in the inner coil is twice as large as that in the outer one and their directions are the same. As a result of the increasing currents a charged particle, initially at rest between the solenoids, starts moving along a circular trajectory (see figure). What is the radius r of the circle?

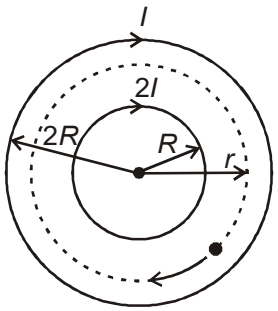


Fig. 36

37. Charge Q is uniformly distributed on a thin insulating ring of mass m which is initially at rest. To what angular velocity will the ring be accelerated when a magnetic field B, perpendicular to the plane of the ring, is switched on?

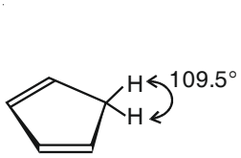
38. In a vacuum chamber a current of 10 A is flowing in a long, straight wire, which has a very high conductivity. Electrons with an initial velocity v_0 start moving perpendicularly towards the wire from a point which is a radial distance r_0 away from the wire. Given that they cannot approach any closer to the wire than $\frac{r_0}{2}$, determine v_0. Ignore the effect of the Earth's magnetic field.

Book Crunch-3

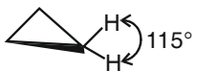
Chemistry

Organic Chemistry

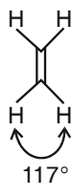
1. Selected bond angles for six hydrocarbons ae shown below. Arrange these hydrocarbons according to their pK_a values, from the lowest to the highest.



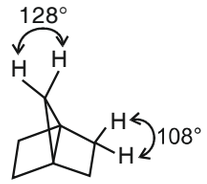
(I)



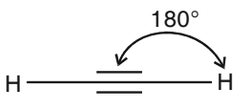
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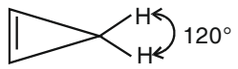
(III)



(IV)



(V)



(VI)

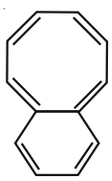
(1) V < I < VI < II < III < IV

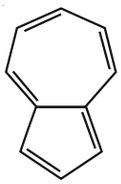
(2) IV < I < II < III < V < VI

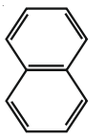
(3) II < IV < I < VI < V < III

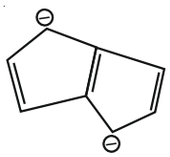
(4) I < V < IV , III < II < VI

2. Which of the following is most polar?

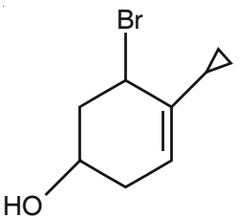
(1) 

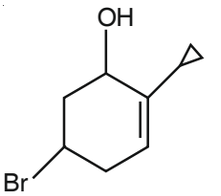
(2) 

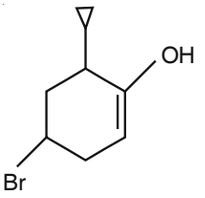
(3) 

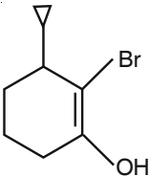
(4) 

3. 5-Bromo-2-cyclopropyl cyclohex-2-enol have correct structure is

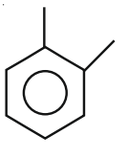
(1) 

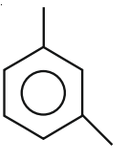
(2) 

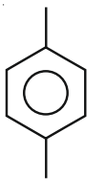
(3) 

(4) 

4. Among the xylenes, which is thermodynamically most stable?

(1) 

(2) 

(3) 

(4) All are equally stable

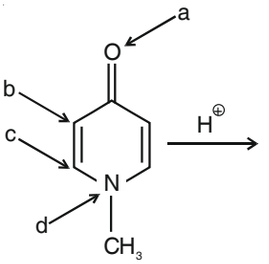
5. Correct order of basic strength of given amines is :

(a) $\underset{2^\circ}{Me_2NH} > \underset{1^\circ}{MeNH_2} > \underset{3^\circ}{Me_3N} > NH_3$ (Protic solvent)

(b) $\underset{2^\circ}{Et_2NH} > \underset{3^\circ}{Et_3H} > \underset{1^\circ}{EtNH_2} > NH_3$ (Protic solvent)

(c) $Me_3N > Me_2NH > Me - NH_2 > NH_3$ (Gas phase)

(d) All are correct

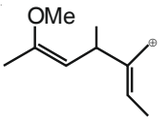
6. 

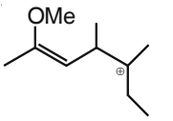
Identify the site, where attack of H^+ is most favourable.

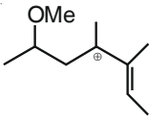
(1) a (2) b

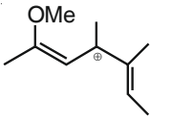
(3) c (4) d

7. Which of the following is the most stabilized carbocation?

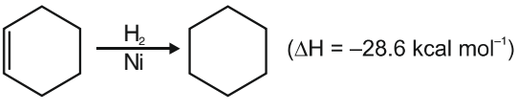
(1) 

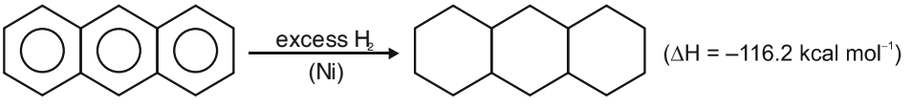
(2) 

(3) 

(4) 

8. Use the following data to answer the question below.





Anthracene

Calculate the resonance energy of anthracene.

(1) 84 kcal/mol

(2) 100 kcal/mol

(3) 110 kcal/mol

(4) 116 kcal/mol

9. Which of the following is not resonating structure of each other?

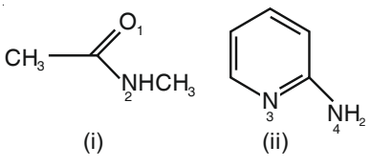
(1) $CH_3 - N = C = S$ and $CH_3 - S - C \equiv N$

(2) $CH_3 - \overset{+}{C} = O$ and $CH_3 - C \equiv \overset{+}{O}$

(3) $CH_3 - \overset{O}{\overset{\|}{C}} - OH$ and $CH_3 - \overset{O^-}{\overset{|}{C}} = \overset{+}{O} - H$

(4) $CH_2 = CH - C \equiv N$ and $\overset{+}{C}H_2 - CH = C = N^-$

10. The preferred sites of protonation in the following compounds are



(i) (ii)

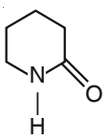
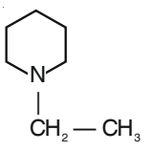
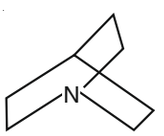
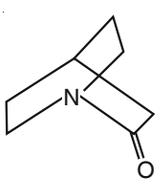
(1) 1 and 3

(2) 2 and 4

(3) 1 and 4

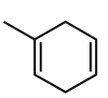
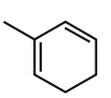
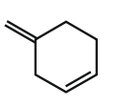
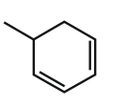
(4) 2 and 3

11. Match the column (I) and (II).

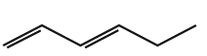
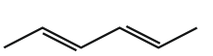
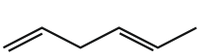
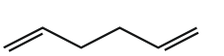
Column I	**Column II**
Molecule	**pK_a of Conjugate acid**
(1) 	(P) 0.8
(2) 	(Q) 5.33
(3) 	(R) 10.65
(4) 	(S) 10.95

12. Select the most stable structure in each of the following.

Part (I)

(1)  (2)  (3)  (4) 

Part (II)

(1)  (2)  (3)  (4) 

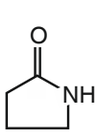
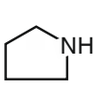
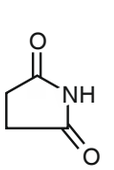
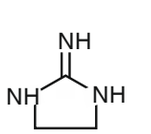
Part (III)

(1) $H_2C = CH - CH = CH - CH_3$ (2) $H_2C = C = CH - CH_2 - CH_3$

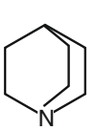
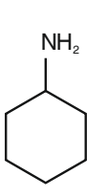
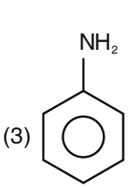
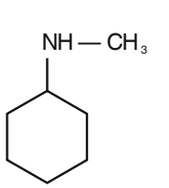
(3) $H_3C - CH = C = CH - CH_3$ (4) $H_2C = CH - CH_2 - CH = CH_2$

13. Identify the most basic compound in the following.

Part (I)

(1)  (2)  (3)  (4) 

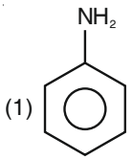
Part (II)

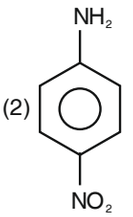
(1)  (2)  (3)  (4) 

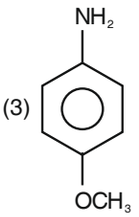
Part (III)

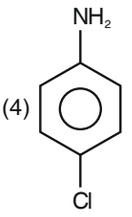
(1) NaOH
(2) $NaOOCCH_3$
(3) $NaOCH_3$
(4) $NaNH_2$

Part (IV)

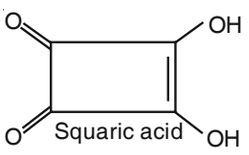
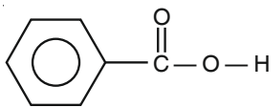
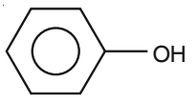
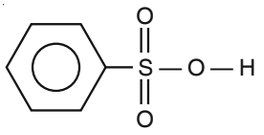
(1) 

(2) 

(3) 

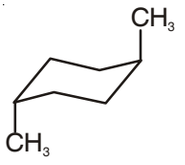
(4) 

14. Match the column I and II.

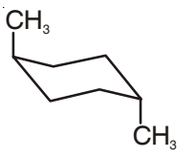
Column II	**Column I**
(1) $NaHCO_3$ will react with	(P)  Squaric acid
(2) Na will react with	(Q) 
(3) NaOH will react with	(R) 
(4) $NaNH_2$ will react with	(S) 

15. The stable form of trans-1, 4-dimethylcyclohexane is represented as :

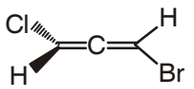
(1) 

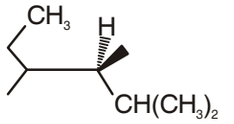
(2) 

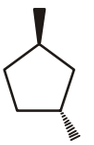
(3) 

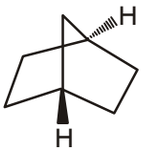
(4) 

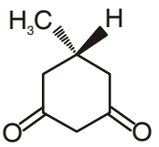
16. Which of the following molecules is (are) chiral?

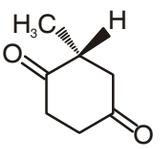

(I)


(II)


(III)


(IV)


(V)


(VI)

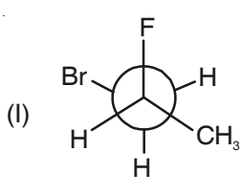
(1) I and II

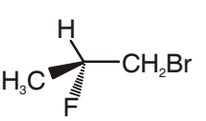
(2) III and IV

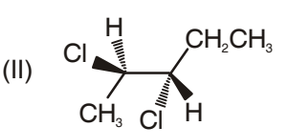
(3) II, IV and VI

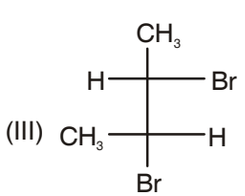
(4) I, II, III and VI

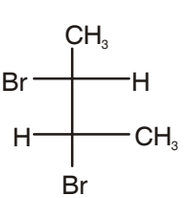
17. Indicate whether each of the following pairs are identical, or ?


(I)



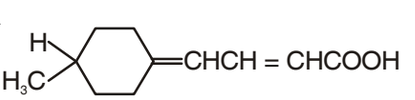

(II)


(III)



	I	II	III
(1)	Enantiomers	Diastereomers	Enantiomers
(2)	Identical	Enantiomers	Enantiomers
(3)	Enantiomers	Diastereomers	Identical
(4)	Enantiomers	Identical	Identical

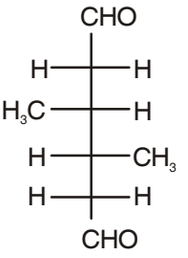
18. How many isomers are possible for the following molecule?

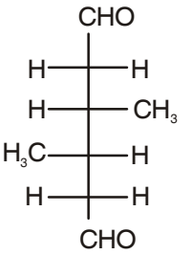


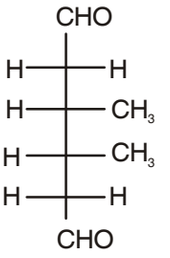
(1) 1 (2) 2

(3) 3 (4) 4

19. Which of the following statements best describes the stereochemical relationships of compound I, II and III shown below?

(I) 

(II) 

(III) 

(1) All compounds are chiral

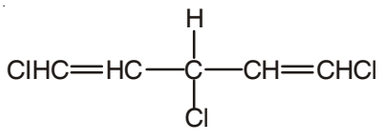
(2) None of the compounds is chiral

(3) I and II are meso compounds

(4) I and II are diastereomers, and III is a meso compound

(5) I and II are chiral

20. How many different stereoisomers are possible for the following compound ?

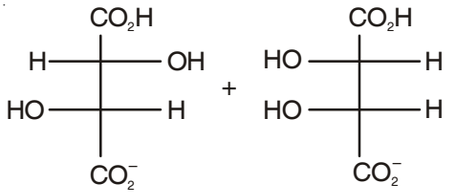


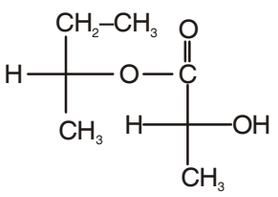
(1) 1 (2) 2

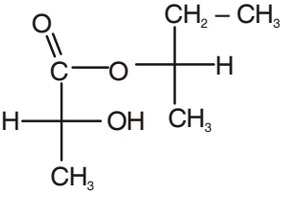
(3) 3 (4) 4

21. Which of the following pair represent pair of diastereomers?

(1) Meso tartaric acid and (*l*) tartaric acid

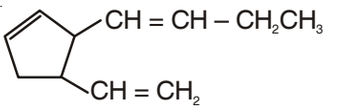
(2) 

(3) 

and 

(4) All of these

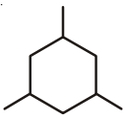
22. Total number of stereoisomers possible for following compound is



(1) 8 (2) 16

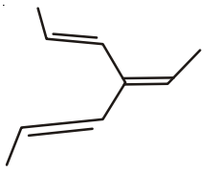
(3) 32 (4) 64

23. Total number of stereoisomer formed by the given compound is



(1) 2 (2) 3

(3) 4 (4) 8

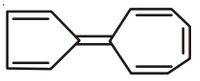
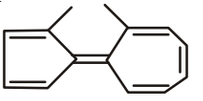
24. 

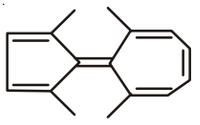
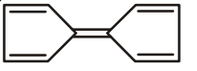
How many geometrical isomers are possible for the above compound?

(1) 3 (2) 4

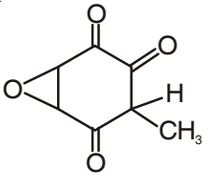
(3) 6 (4) 8

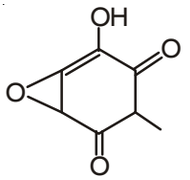
25. Which of following compound will rotate the plane polarized light at room temperature?

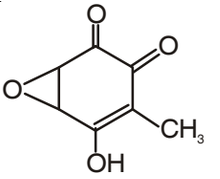
(1)  (2) 

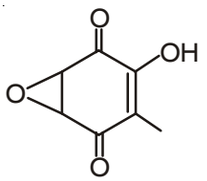
(3)  (4) 

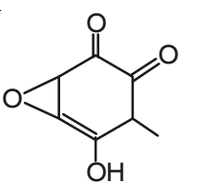
26. Identify most stable enol form of terreic acid



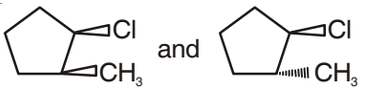
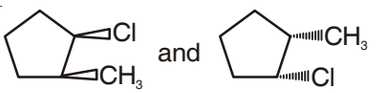
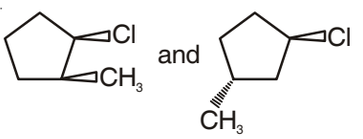
(1) 

(2) 

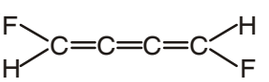
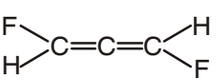
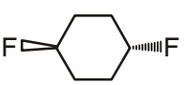
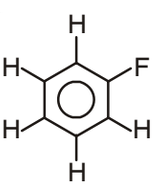
(3) 

(4) 

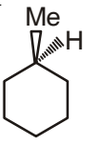
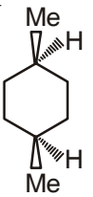
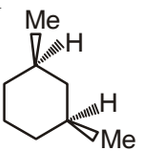
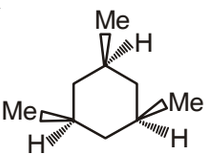
27. Match the column (I) and (II).

Column-I **Molecules**	**Column-II** **Relationship**
(1) 	(P) Identical
(2) 	(Q) Enantiomer
(3) 	(R) Diastereomer
(4) 	(S) Structural Isomerism

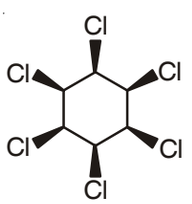
28. Match the column (I) and (II).

Column-I **Molecule**	**Column-II** **Property**
(1) 	(P) Polar molecule
(2) 	(Q) Optically active
(3) 	(R) Optically inactive
(4) 	(S) Plane of symmetry

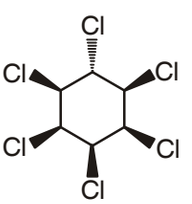
29. Match the column (I) and (II).

Column-I **Molecule**	**Column-II** **Property**
(1) 	(P) Meso compound
(2) 	(Q) Achiral
(3) 	(R) Chiral compound
(4) 	(S) Compound will show geometrical isomerism

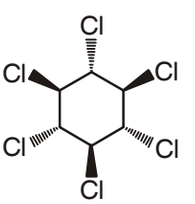
30. Comprehension



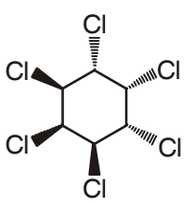
(P)



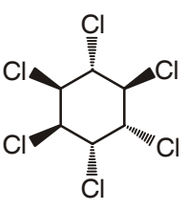
(Q)



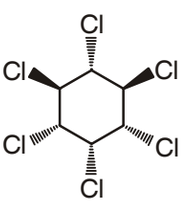
(R)



(S)



(T)



(U)

Consider the given structures and answer A and B.

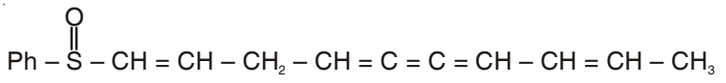
A. Which of the compound is optically active ?

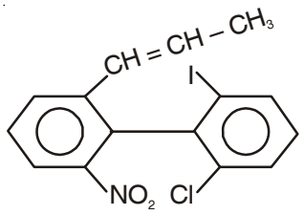
(1) P (2) R (3) S (4) T

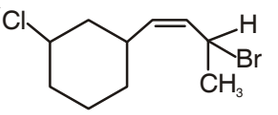
B. Which of the isomer is most stable?

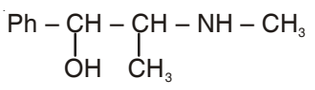
(1) R (2) S (3) T (4) U

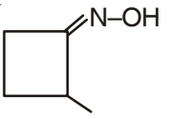
31. Find total number of stereoisomers for each compound given below.

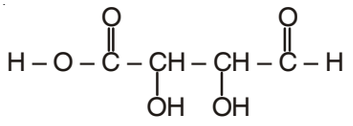
(1) 

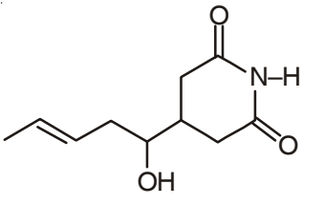
(2) 

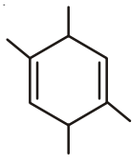
(3) 

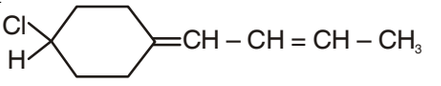
(4) 

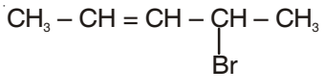
(5) 

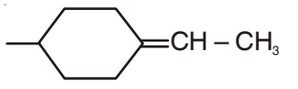
(6) 

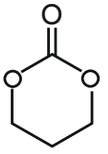
(7) 

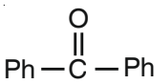
(8) 

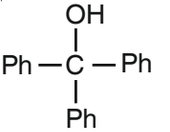
(9) 

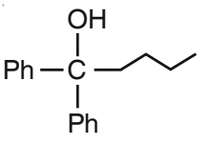
(10) 

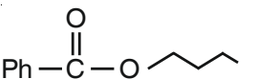
(11) 

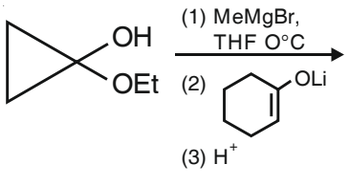
32. PhMgBr (excess) +  $\xrightarrow{H^+}$ (A). Product (A) is

(1) 

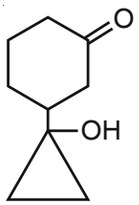
(2) 

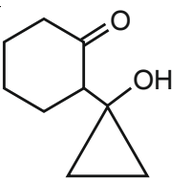
(3) 

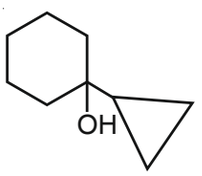
(4) 

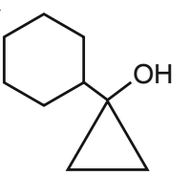
33. 

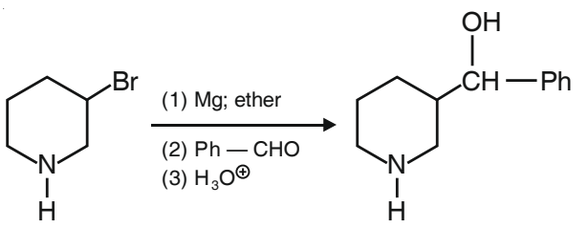
(A); Product (A) in this sequence is

(1) 

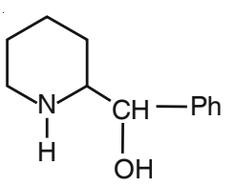
(2) 

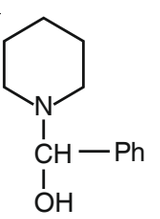
(3) 

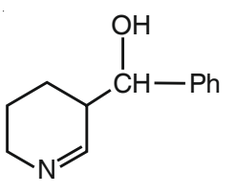
(4) 

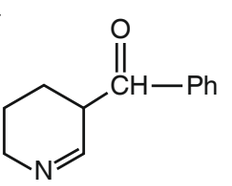
34. 

The given product can not be obtained in the above reaction. Identify the correct product obtained.

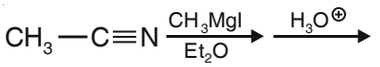
(1) 

(2) 

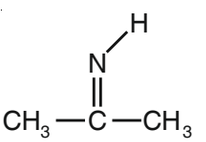
(3) 

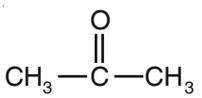
(4) 

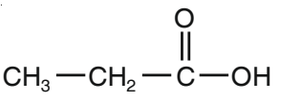
35. What is the major product of the following reaction?



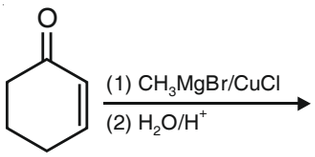
(1) $CH_3-CH_2-NH-CH_3$

(2) 

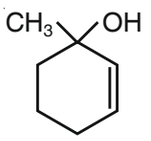
(3) 

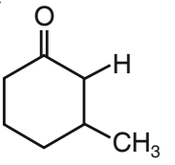
(4) 

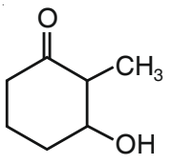
36. In the reaction sequence

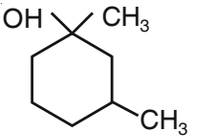


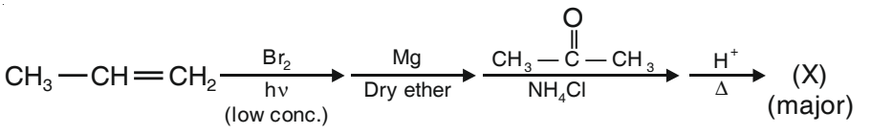
(X), product (X) will be

(1) 

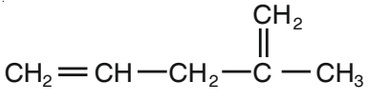
(2) 

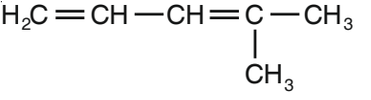
(3) 

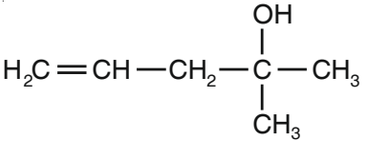
(4) 

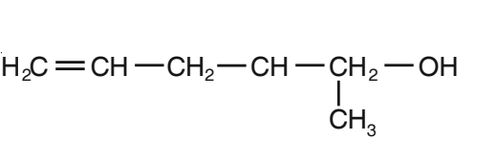
37. 

End product (X) of the above reaction is

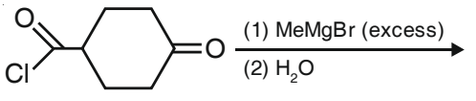
(1) 

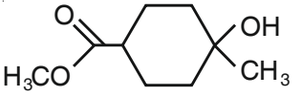
(2) 

(3) 

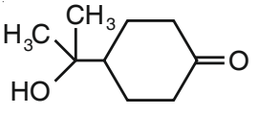
(4) 

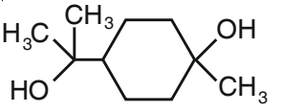
38. Give the expected product of the following reaction.



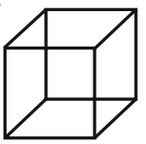
(1) 

(2) 

(3) 

(4) 

39. Double bond equivalent of cubane is



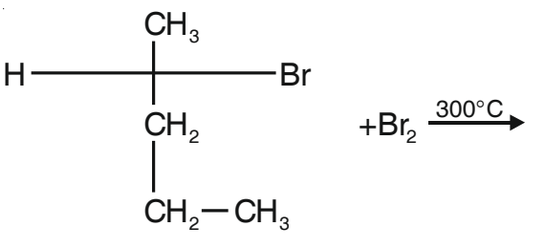
Cubane

(1) 4

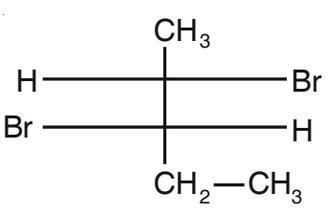
(2) 5

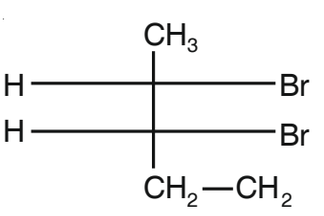
(3) 6

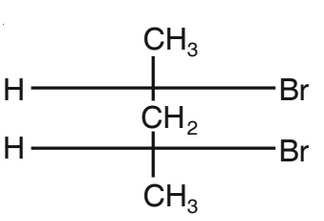
(4) 7

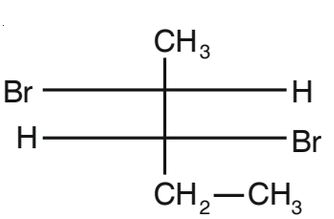
40. 

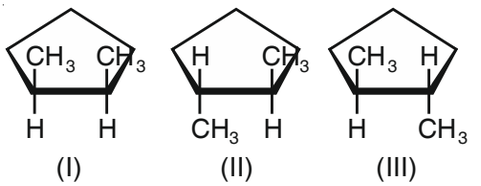
Which of the following compound will not be obtained as a product in the above reaction?

(1) 

(2) 

(3) 

(4) 

41. 

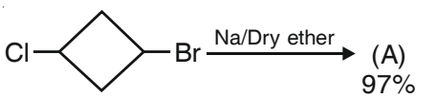
Among the structures given, select the enantiomers.

(1) I and II

(2) I and III

(3) II and III

(4) I, II and III

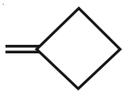
42. 

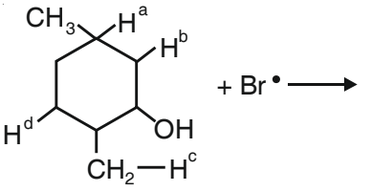
Product (A) of above reaction is

(1) 

(2) 

(3) 

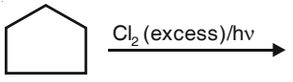
(4) 

43. 

$Br^{\bullet}$ will abstract which of the hydrogen most readily?

(1) a (2) b

(3) c (4) d

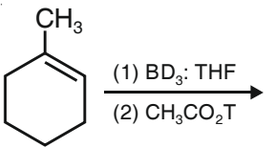
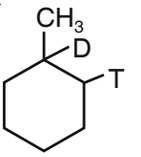
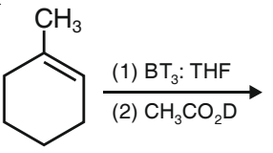
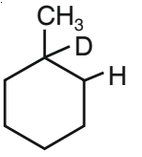
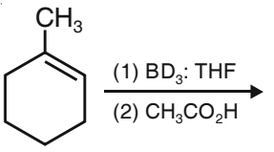
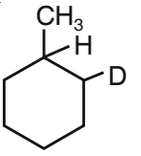
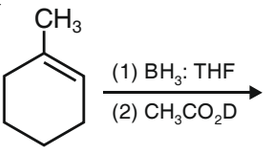
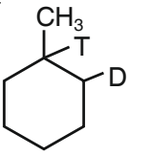
44. 

How many dichloro products are formed in the above reaction (including stereoisomers)?

(1) 5 (2) 6

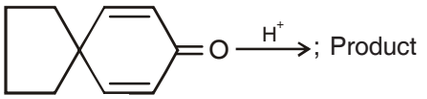
(3) 7 (4) 9

45. Match the column.

Column (I) **Reaction**	**Column (II)** **Product**
(1) 	(P) 
(2) 	(Q) 
(3) 	(R) 
(4) 	(S) 

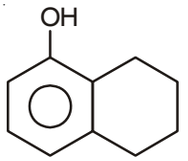
46. Match the column.

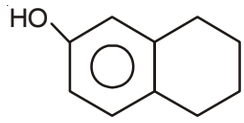
Column (I) **Wurtz reaction**	**Column (II)** **Number of dimerization product**
(1) $CH_3 - Cl \xrightarrow[\text{dry ether}]{Na}$	(P) 5
(2) $CH_3 - Cl + CH_3 - CH_2 - Cl \xrightarrow[\text{dry ether}]{Na}$	(Q) 6
(3) $CH_3 - Cl + CH_3 - CH_2 - Cl + CH_3 - CH_2 - CH_2 - Cl \xrightarrow[\text{dry ether}]{Na}$	(R) 3
(4) $H_2C = CH - CH = CH - CH_2 - Cl + CH_3 - CH_2 - Cl \xrightarrow[\text{dry ether}]{Na}$	(S) 1

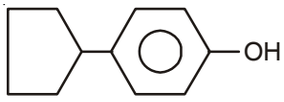
47. 

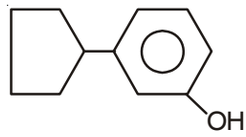
; Product

Major product of the reaction is :

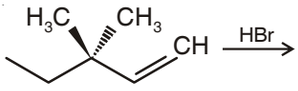
(1) 

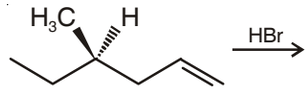
(2) 

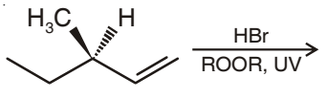
(3) 

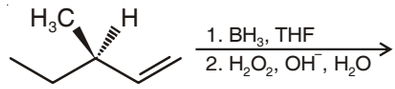
(4) 

48. Which of the following reactions results in the formation of a pair of diastereomers?

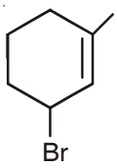
(1) 

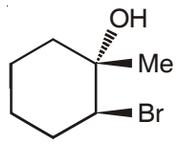
(2) 

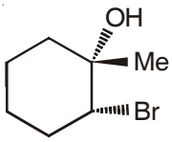
(3) 

(4) 

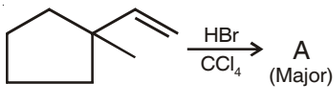
49. Which of the following is the best stereochemical representation when reaction between 1-methylcyclohexene and NBS react in aqueous dimethyl sulfoxide?

(1) 

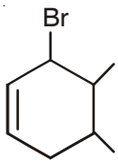
(2) 

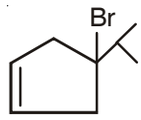
(3) 

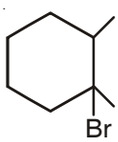
(4) None of these

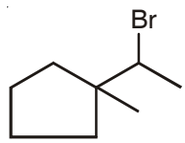
50. 

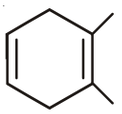
Product (A) is :

(1) 

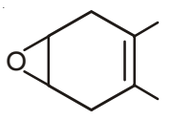
(2) 

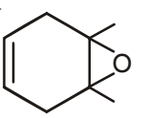
(3) 

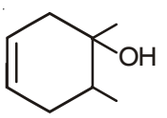
(4) 

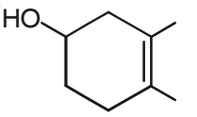
51.  $\xrightarrow{\text{MCPBA}}$ (A); MCPBA $\rightarrow$ metachloroperbenzoic acid

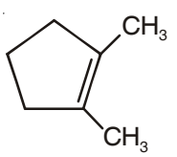
Product (A) of the above reaction is :

(1) 

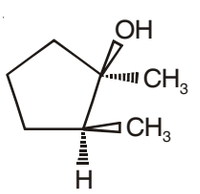
(2) 

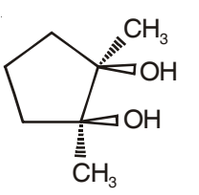
(3) 

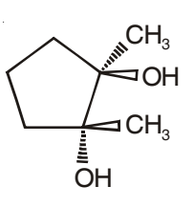
(4) 

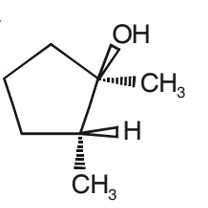
52.  $\xrightarrow[2.\ NaHSO_3]{1.\ OsO_4}$ Product

Product of the reaction is :

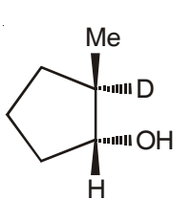
(1) 

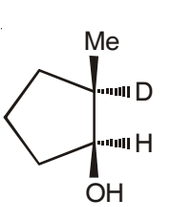
(2) 

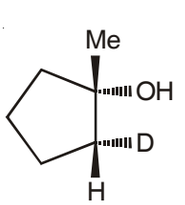
(3) 

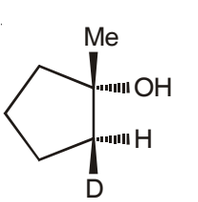
(4) 

53. Hydroboration of 1-methylcyclopentene using B_2D_6, followed by treatment with alkaline hydrogen peroxide, gives

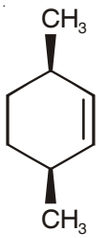
(1) 

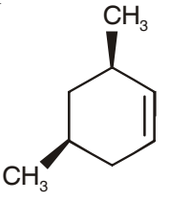
(2) 

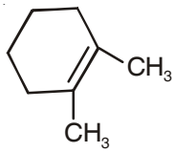
(3) 

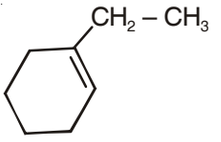
(4) 

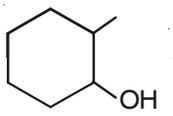
54. An optically active compound A with molecular formula C_8H_{14} undergoes catalytic hydrogenation to give meso compound, the structure of (A) is :

(1) 

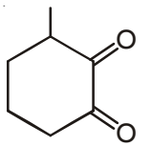
(2) 

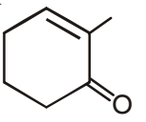
(3) 

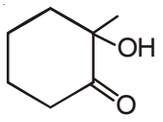
(4) 

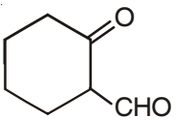
55.  $\xrightarrow[\Delta]{H^+}$ (A) $\xrightarrow[KMnO_4]{\text{cold dil.}}$ (B) $\xrightarrow{CrO_3}$ (C)

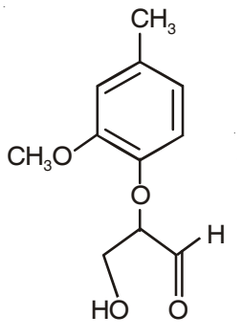
Product (C) of the reaction is :

(1) 

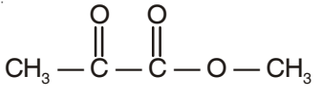
(2) 

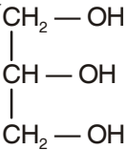
(3) 

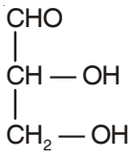
(4) 

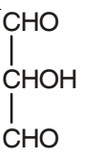
56.  $\xrightarrow{NaBH_4}$ A $\xrightarrow{O_3}$ B $\xrightarrow{H_2O}$ (C) (one of the product)

Identify the product (C) :

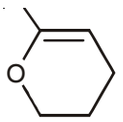
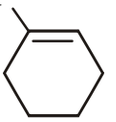
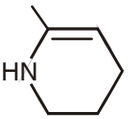
(1) 

(2) 

(3) 

(4) 

57. Rank the following in the increasing order of rate of reaction with HBr.

(P)  (Q)  (R) 

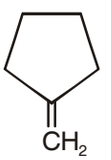
(1) R > P > Q

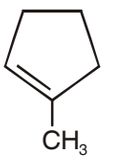
(2) R > Q > P

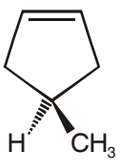
(3) P > R > Q

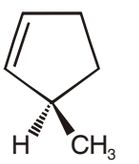
(4) P > Q > R

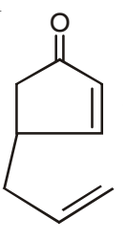
58. Which of the following compound give diastereomers when treated with Br_2 in CCl_4?

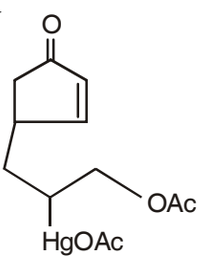
(1) 
Methylcyclopentane

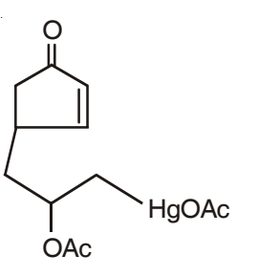
(2) 
1-Methylcyclopentene

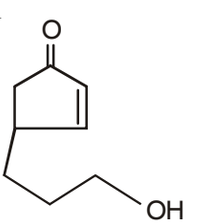
(3) 
3-Methylcyclopentene

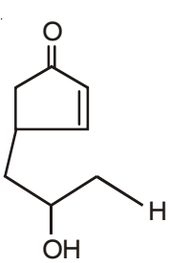
(4) 
4-Methylcyclopentene

59.  $\xrightarrow[AcOH]{Hg(OAc)_2}$ X (major); product (X) is :

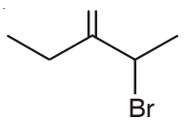
(1) 

(2) 

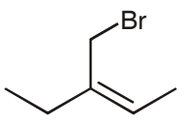
(3) 

(4) 

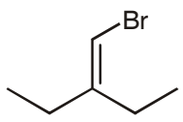
60. Which of the following compounds gives the same carbocation on ionization?



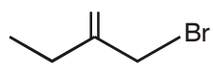
1



2



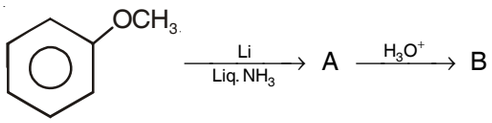
3



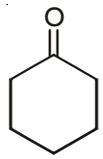
4

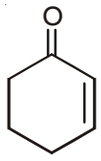
(1) 1 and 3 (2) 2 and 4

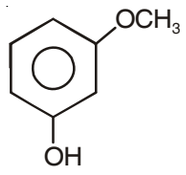
(3) 1 and 2 (4) 1 and 4

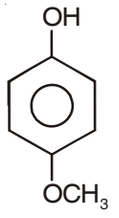
61. 

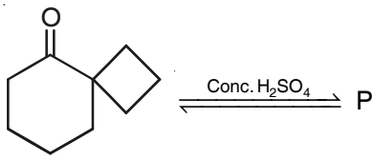
Product (B) in the above reaction is :

(1) 

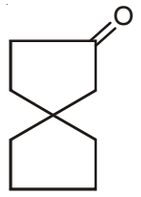
(2) 

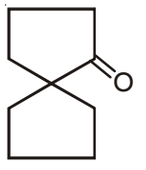
(3) 

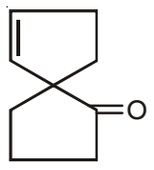
(4) 

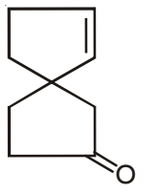
62. 

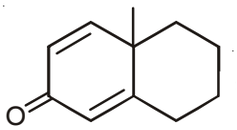
Identify product (P).

(1) 

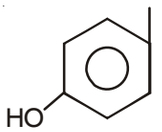
(2) 

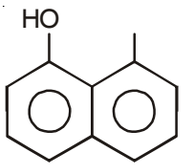
(3) 

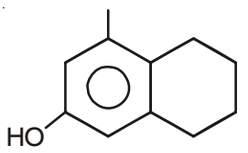
(4) 

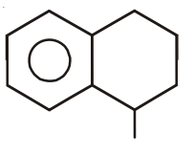
63.  $\overset{H^+}{\rightleftharpoons}$ Product

Predict the major Product :

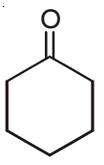
(1) 

(2) 

(3) 

(4) 

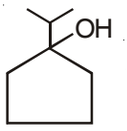
64. In which of the following reactions, two products will be formed other than phosphonium ylide ($POPh_3$)?

(1)  $+ Ph_3P = CH_2 \longrightarrow$

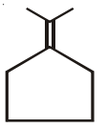
(2) $CH_3CHO + Ph_3P = \underset{\displaystyle CH_3}{\overset{\displaystyle |}{C}} - CH_3 \longrightarrow$

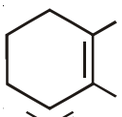
(3) $Ph - \overset{\displaystyle O}{\overset{\displaystyle ||}{C}} - H + Ph_3P = CH - Ph \longrightarrow$

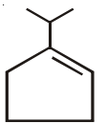
(4) $H - \overset{\displaystyle O}{\overset{\displaystyle ||}{C}} - H + Ph_3P = CH - CH_3 \longrightarrow$

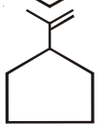
65.  $\xrightarrow[\Delta]{H^+}$ (A) (major)

Product (A) is :

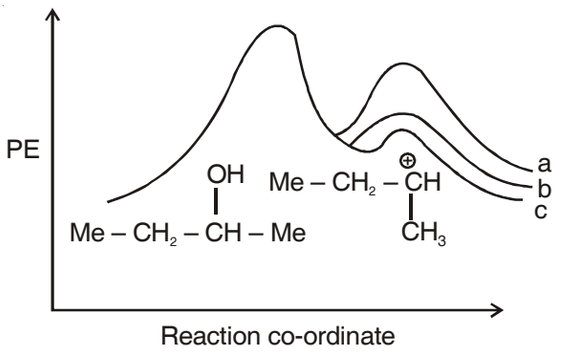
(1) 

(2) 

(3) 

(4) 

66. Energy profile diagram for dehydration of 2-butanol using conc. H_2SO_4 is given below :



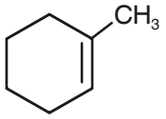
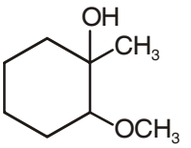
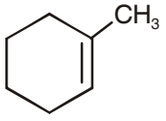
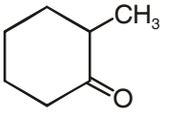
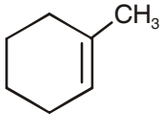
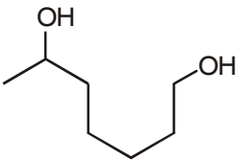
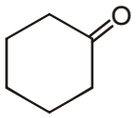
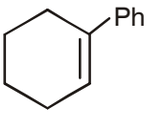
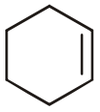
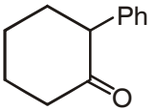
Product (b) of above reaction is :

(1) 1-butene (2) *cis*-2-butene

(3) *trans*-2-butene (4) *iso*-butene

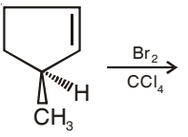
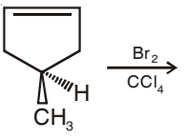
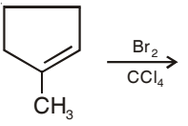
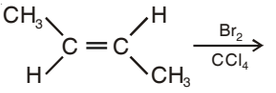
67. In each reaction box write a single letter designating the best reagent and condition selected from the list at bottom of the page.

(F.S. → first step, S.S. → second step, T.S. → third step)

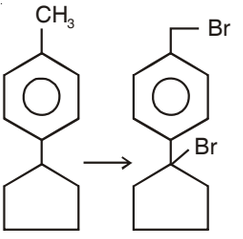
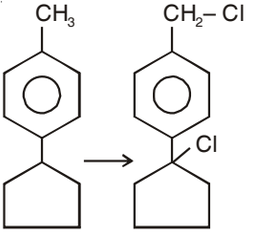
Reaction	Reactant	Options	Product
1.		F.S. □ → S.S. □	
2.		F.S. □ → S.S. □ T.S. □	
3.		F.S. □ → S.S. □	
4.		F.S. □ → S.S. □	
5.		F.S. □ → S.S. □ T.S. □	

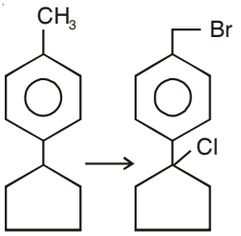
(A) $NaBH_4$/alcohol	(B) Ph – CO_3H/CH_2Cl_2	(C) PCC	(D) CH_3ONa/CH_3OH
(E) B_2H_6 in THF	(F) H_2O_2/aq. NaOH	(G) H_3PO_4 & heat	(H) $AlCl_3/C_6H_6$
(I) O_3 in CH_2Cl_2	(J) Br_2 in CH_2Cl_2	(K) 20% KOH & heat	(L) Ph – Li/ether

68. Match the Column-I with Column-II and with Column-III (Matrix).

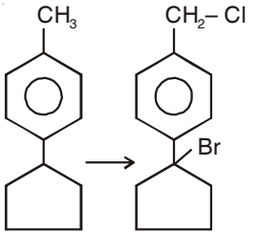
Column I Reaction	**Column II** Nature of product formed	**Column III** Number of chiral center present in product. (consider only one isomer in case of racemic mixture or diastereomer)
(1) 	(P) Racemic mixture	(W) 0
(2) 	(Q) Meso	(X) 1
(3) 	(R) Diastereomer	(Y) 2
(4) 	(S) Vicinal dihalide	(Z) 3

69. Match the Column-I and II :

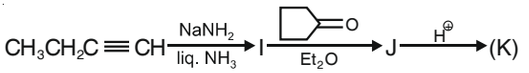
Column I **Conversion**	**Column II** **Reagent**
(1) 	(P) $SO_2Cl_2/h\nu$ (2 equivalent)
(2) 	(Q) NSB (2 equivalent)

(3) 

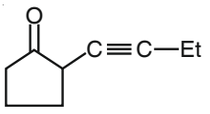
(R) NSB then $SO_2C_2I/h\nu$

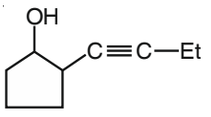
(4) 

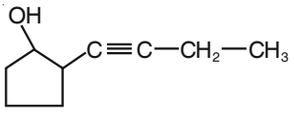
(S) $SO_2Cl_2/h\nu$ then NBS

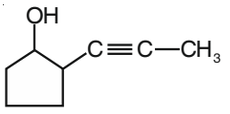
70. 

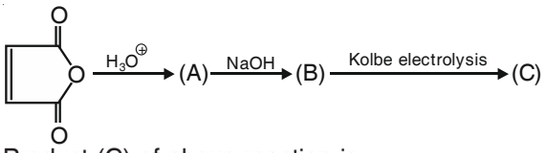
Product (K) of the above reaction is

(1) 

(2) 

(3) 

(4) 

71. 

Product (C) of above reaction is

(1) $H_2C=CH_2$

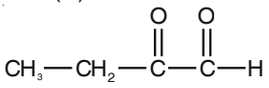
(2) $CH_3-C\equiv C-CH_3$

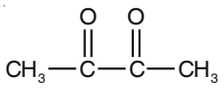
(3) $HC\equiv CH$

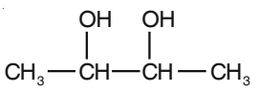
(4) $CH_3-CH=CH-CH_3$

72. $CH_3-C\equiv C-CH_3 \xrightarrow{\text{Cold } KMnO_4} (A)$

Product (A) is

(1) 

(2) 

(3) 

(4) $O=CH-CH_2-CH_2CH=O$

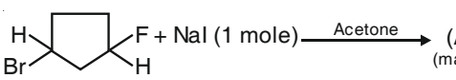
73. The best sequence of reactions to prepare 2-heptanone is

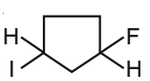
(1) propyne $\xrightarrow{NaNH_2}$ X $\xrightarrow{n\text{–}C_4H_9Br}$ Y $\xrightarrow[H_2SO_4]{H_2O,\ Hg^{2+}}$

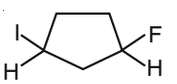
(2) ethyne $\xrightarrow{NaNH_2}$ X $\xrightarrow{n\text{–}C_5H_{11}Br}$ Y $\xrightarrow[H_2SO_4]{H_2O,\ Hg^{2+}}$

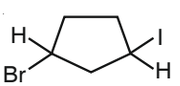
(3) 1–hexyne $\xrightarrow{NaNH_2}$ X $\xrightarrow{CH_3Br}$ Y $\xrightarrow[H_2SO_4]{H_2O,\ Hg^{2+}}$

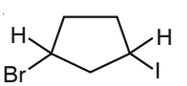
(4) 1–pentyne $\xrightarrow{NaNH_2}$ X $\xrightarrow{C_2H_5Br}$ Y $\xrightarrow[H_2SO_4]{H_2O,\ Hg^{2+}}$

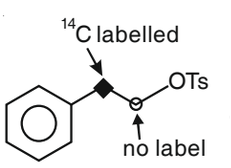
74.  ; Major product of this reaction is

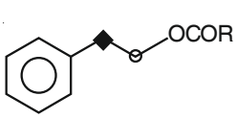
(1) 

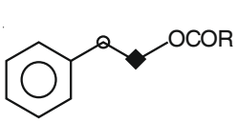
(2) 

(3) 

(4) 

75.  product; product of this reaction is

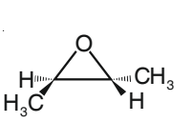
(1) 

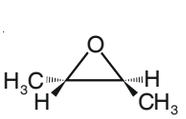
(2) 

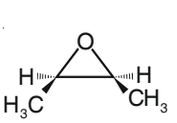
(3) Both (1) and (2)

(4) None of these

76. (Z)-2-Butene reacts with Br_2/H_2O. The resulting bromohydrin when treated with methoxide in methanol undergoes an intramolecular S_N2 reaction. Taking into consideration the stereochemical consequences of the reaction mechanism involved, choose the final product(s) of these transformations.

(I) 

(II) 

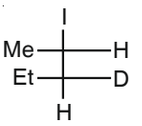
(III) 

(1) (I) only

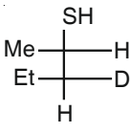
(2) (II) only

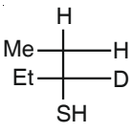
(3) (III) only

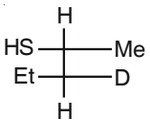
(4) Equal amounts of (I) and (II)

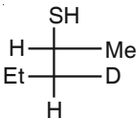
77. 

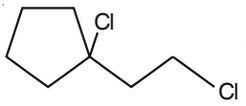
KSH → Product; Identify the product.

(1) 

(2) 

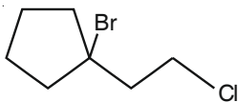
(3) 

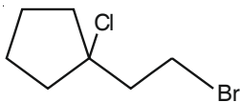
(4) 

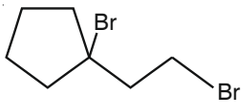
78. 

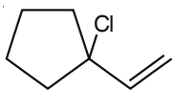
LiBr/DMSO, S_N2 conditions → Major product (X)

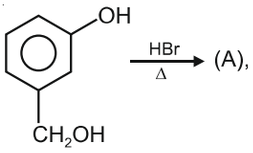
The product X is

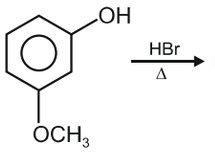
(1) 

(2) 

(3) 

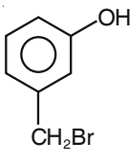
(4) 

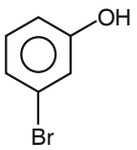
79. 

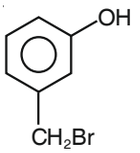
HBr, Δ → (A),


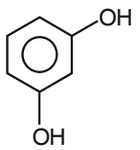
HBr, Δ → (B),

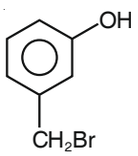
Product (A) and (B) respectively are:

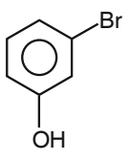
(1) 

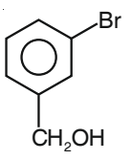
and 

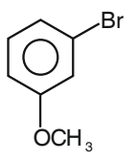
(2) 

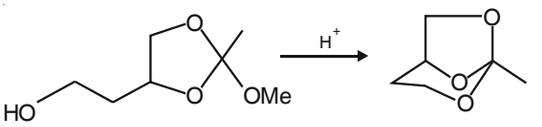
and 

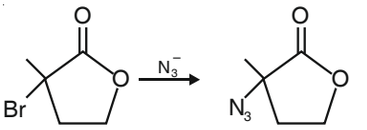
(3) 

and 

(4) 

and 

80.  **Reaction 1**

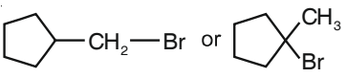
 **Reaction 2**

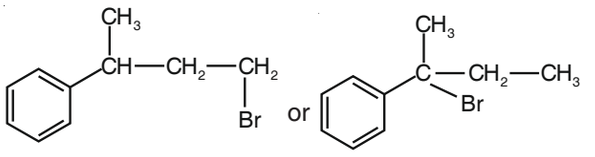
Types of mechanism followed by reaction 1 and reaction 2 respectively

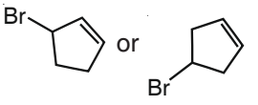
(1) S_N1, S_N1 (2) S_N2, S_N2

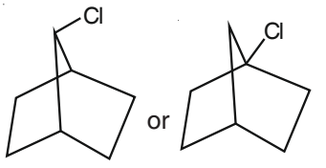
(3) S_N1, S_N2 (4) S_N2, S_N1

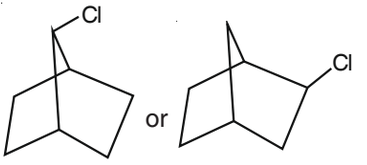
81. Encircle whichever of the following :

(1) undergoes an S_N2 reaction more rapidly : 

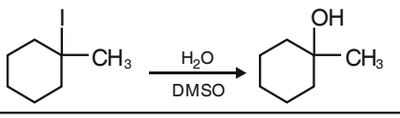
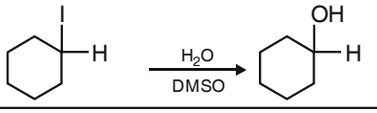
(2) undergoes an E_1 reaction more rapidly : 

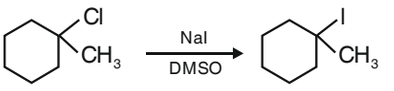
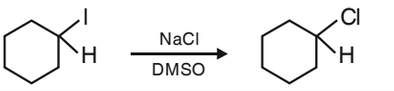
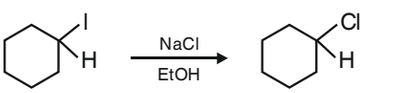
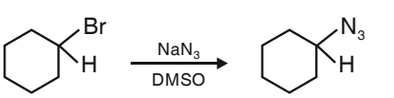
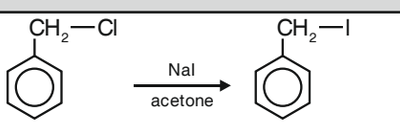
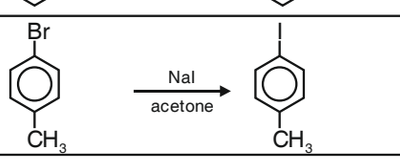
(3) undergoes an S_N1 reaction more rapidly : 

(4) undergoes an S_N2 reaction more rapidly : 

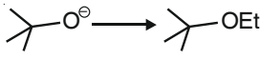
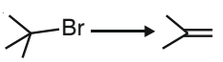
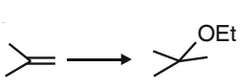
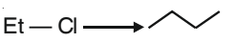
(5) undergoes an E_2 reaction more rapidly : 

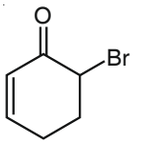
82. Select which reaction from the following reaction pairs will occur faster.

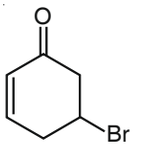
Part-1	
Reaction A	
Reaction B	

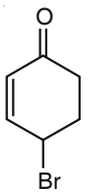
Part-2	
Reaction C	
Reaction D	
Part-3	
Reaction E	
Reaction F	
Part-4	
Reaction G	
Reaction H	
Part-5	
Reaction I	
Reaction J	

83.

Column I **(Reaction sequence)**	**Column II** **(Reagent required)**
(1) 	(P) $EtO^{\ominus}$
(2) 	(Q) EtBr
(3) 	(R) $EtOH/H^{\oplus}$
(4) 	(S) Et—Cl/Na ether

84. Reaction (1)  ; $\xrightarrow{\text{alc. KOH}}$ (A) (major)

Reaction (2)  $\xrightarrow{\text{alc. KOH}}$ (B) (major)

Reaction (3)  $\xrightarrow{\text{alc. KOH}}$ (C) (major)

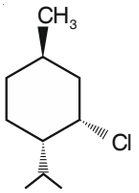
Product obtained in above reactions (1), (2) and (3) is

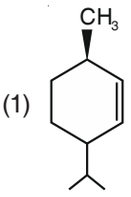
(1) A = B but C is different

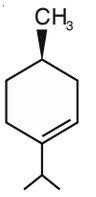
(2) A = C, but B is different

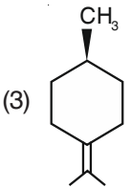
(3) B = C, but A is different

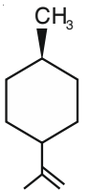
(4) A = B = C all product are identical

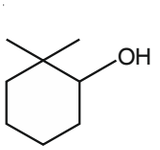
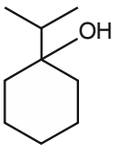
85.  $\xrightarrow[(E_2 \text{ reaction})]{\text{NaOEt}}$; Major product of the reaction is :

(1) 

(2) 

(3) 

(4) 

86.  $\xrightarrow[\Delta]{H^{\oplus}}$ (A);  $\xrightarrow[\Delta]{H^{\oplus}}$ (B)

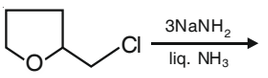
Sum of number of α-hydrogen present it compound A + B is

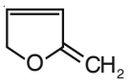
(1) 18

(2) 19

(3) 20

(4) 21

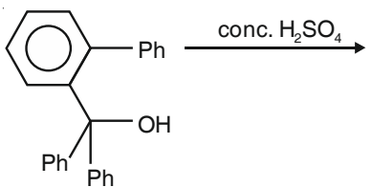
87.  Product; The product is

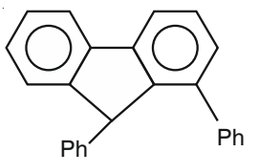
(1) 

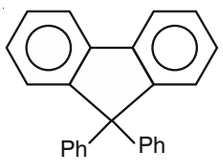
(2) $HC \equiv C-(CH_2)_3ONa$

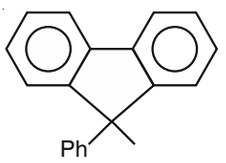
(3) $NaC \equiv C-(CH_2)_3ONa$

(4) $H-C \equiv C-(CH_2)_3OH$

88.  Major product of the reaction is

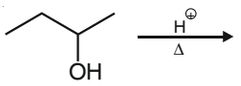
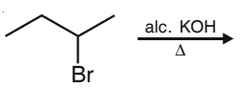
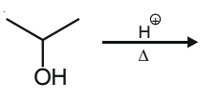
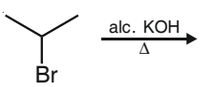
(1) 

(2) 

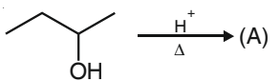
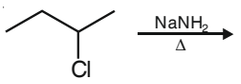
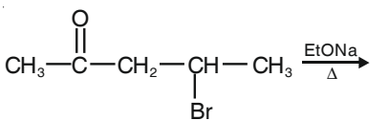
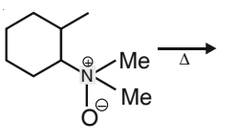
(3) 

(4) None of these

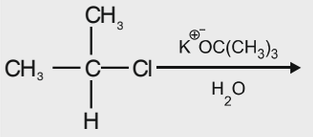
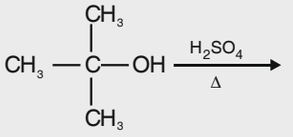
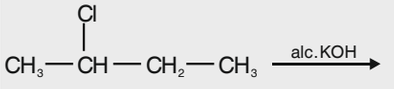
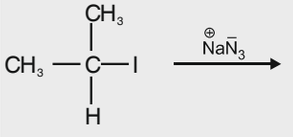
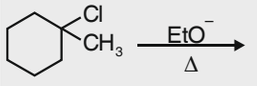
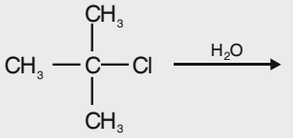
89. Match the column

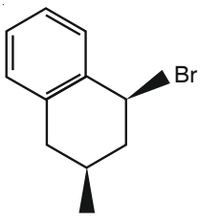
Column I	**Column II**
(1) 	(P) Product are Diastereomers
(2) 	(Q) Carbocation is intermediate
(3) 	(R) 2nd order reaction
(4) 	(S) 1st order reaction

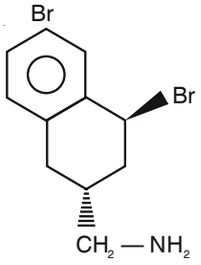
90. Match the column

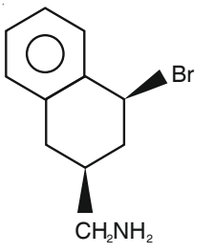
Column I	Column II
(1) 	(P) E_1
(2) 	(Q) E_2
(3) 	(R) Ei (elimination intramolecular)
(4) 	(S) E_{1CB}

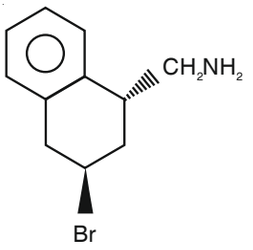
91. Select whether the following reagent combination will result in elimination or substitution reactions leading to the major product.

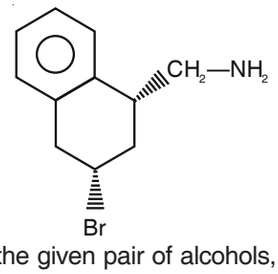
Reaction	Substitution	Elimination
(1) 		
(2) 		
(3) 		
(4) 		
(5) 		
(6) 		

92.  $\xrightarrow[2.\ LiAlH_4]{1.\ One\ eq.\ NaCN}$ (A) ; Product (A) is

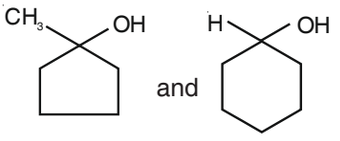
(1) 

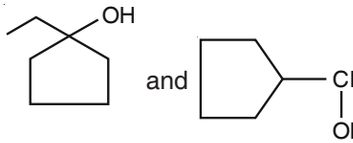
(2) 

(3) 

(4) 

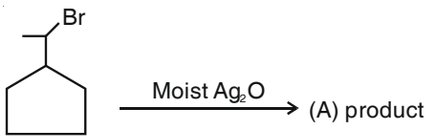
93. In the given pair of alcohols, in which pair second alcohol is more reactive than first towards hydrogen bromide?

(1) 

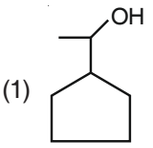
(2) 

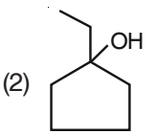
(3) $CH_3-\underset{\underset{OH}{|}}{CH}-CH_2-CH_3$ and $CH_3-CH_2-\underset{\underset{CH_3}{|}}{CH}-CH_2-OH$

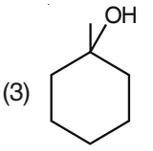
(4) $CH_3-\underset{\underset{OH}{|}}{CH}-CH_2-CH_3$ and $(CH_3)_2\underset{\underset{OH}{|}}{C}-CH_2-CH_3$

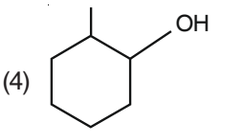
94. 

Major product (A) is

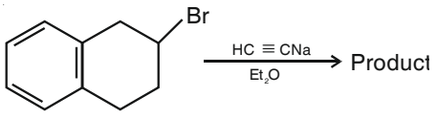
(1) 

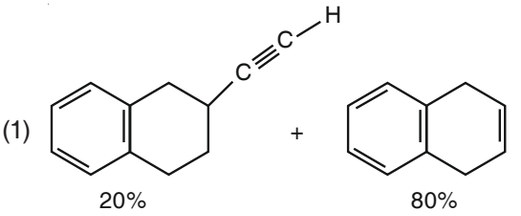
(2) 

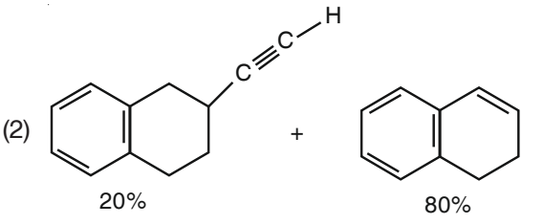
(3) 

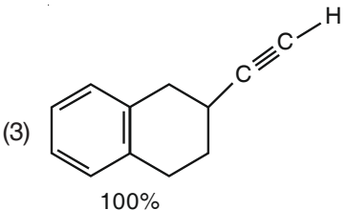
(4) 

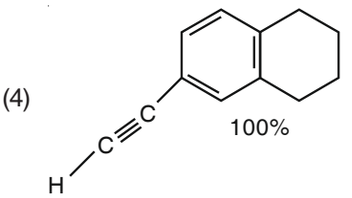
95. What are the products obtained from the following reaction?



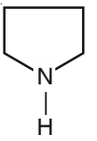
(1) 

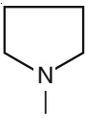
(2) 

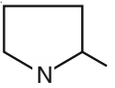
(3) 

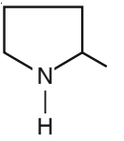
(4) 

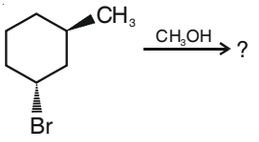
96. $Br — CH_2—(CH_2)_2 — CH_2 — Br + CH_3NH_2 \longrightarrow$ Product of the reaction is

(1) 

(2) 

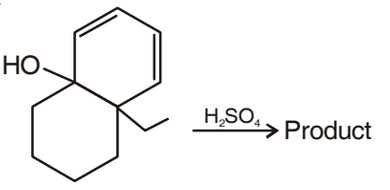
(3) 

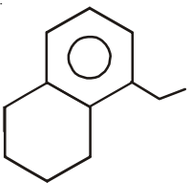
(4) 

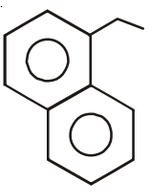
97. 

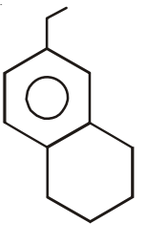
x = total number of substitution and elimination product(s). Find the value of x.

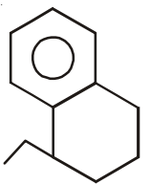
98. Identify the major product



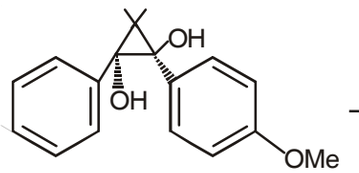
(1) 

(2) 

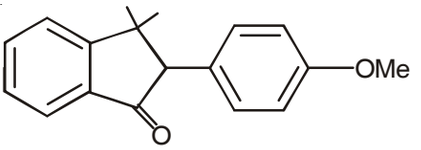
(3) 

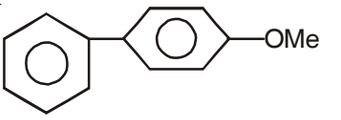
(4) 

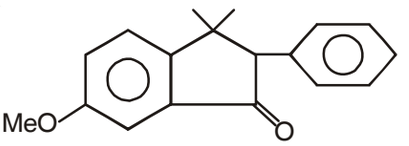
99. Predict the major product of the given reaction.

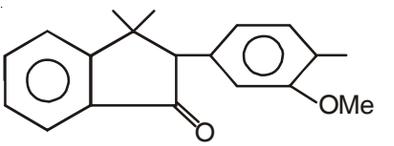


(A); Product (A) is

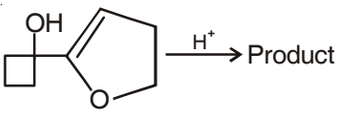
(1) 

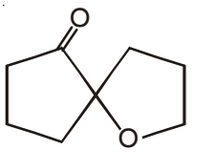
(2) 

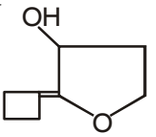
(3) 

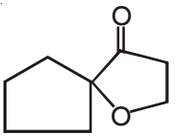
(4) 

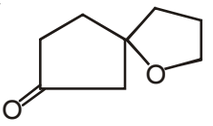
100. Identify the major product,

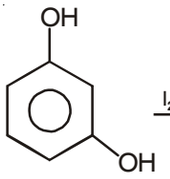


(1) 

(2) 

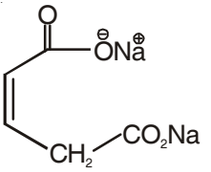
(3) 

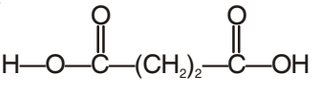
(4) 

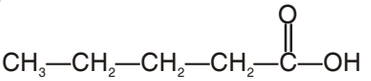
101. 

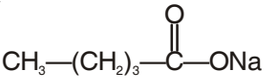
(Resorcinol)

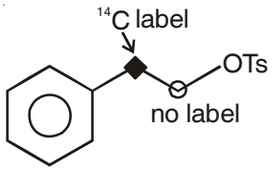
$I_2 + NaOH \rightarrow CHI_3 + (A)$; Compound (A) is
(Iodoform)

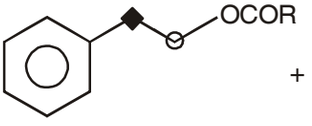
(1) 

(2) 

(3) 

(4) 

102. 

$\xrightarrow{RCO_2H}$; 

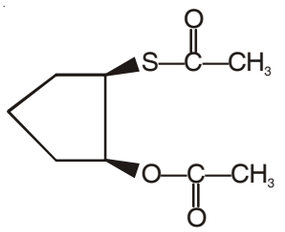
+ 

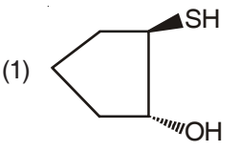
(1) S_N1

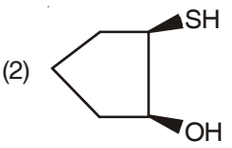
(2) S_N2

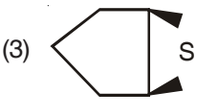
(3) SN — NGP

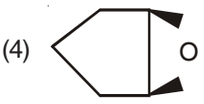
(4) SN — Ar

103.  $\xrightarrow[2.\ H^+]{1.\ HO^-}$ Major Product of the reaction is

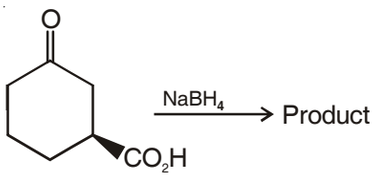
(1) 

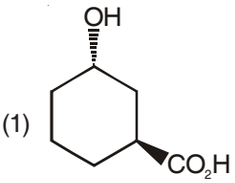
(2) 

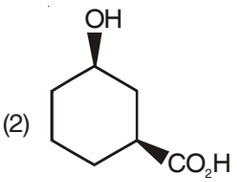
(3) 

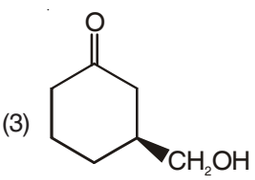
(4) 

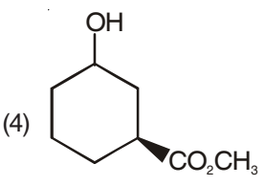
104. What is the major product of the following reaction?



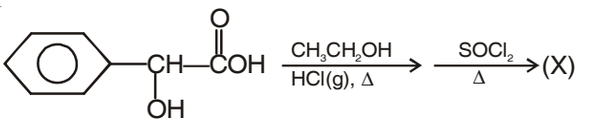
(1) 

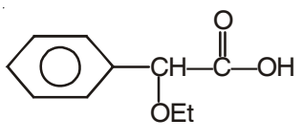
(2) 

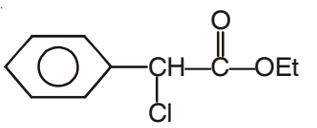
(3) 

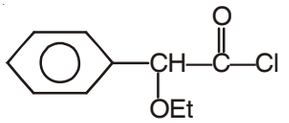
(4) 

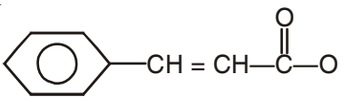
105. Assign the structure of major product (X) of the reaction given below



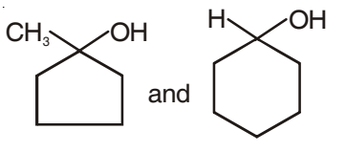
(1) 

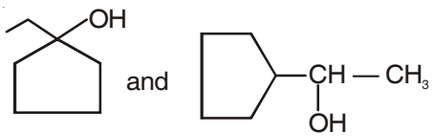
(2) 

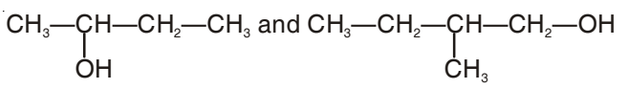
(3) 

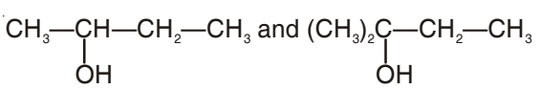
(4) 

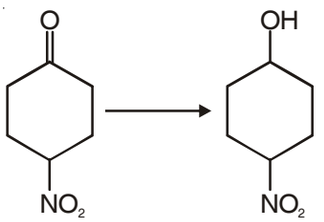
106. In the given pair of alcohol, in which pair second alcohol is more reactive than first towards hydrogen bromide?

(1) 

(2) 

(3) 

(4) 

107. 

Above conversion can be achieved by

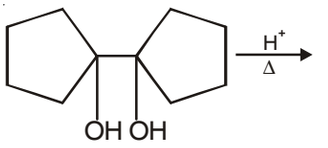
(1) $LiAlH_4$

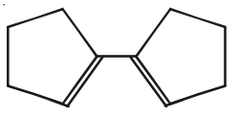
(2) $NaBH_4$

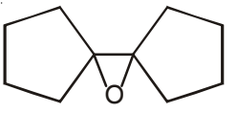
(3) H_2/Ni

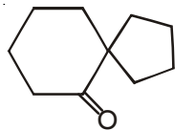
(4) CrO_3

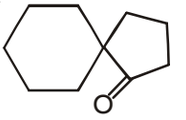
108. The structure of the product formed in the reaction given below is

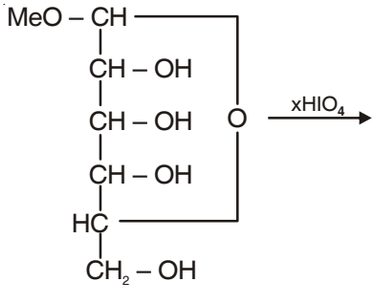


(1) 

(2) 

(3) 

(4) 

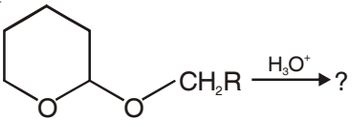
109. 

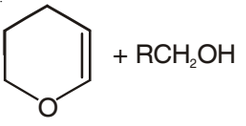
What is the maximum value of (x) ?

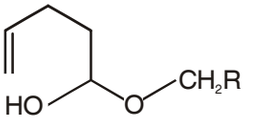
(1) 1 (2) 2

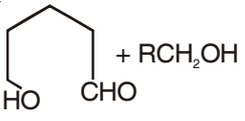
(3) 3 (4) 4

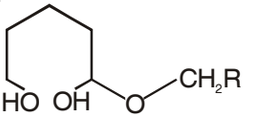
110. The major product formed in the reaction is



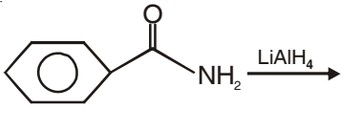
(1) 

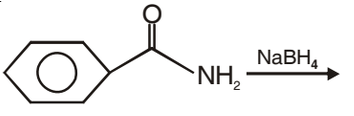
(2) 

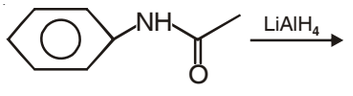
(3) 

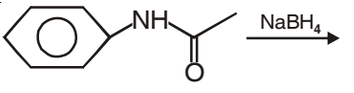
(4) 

111. Find out the reaction in which obtained product give positive isocyanide test.

(1) 

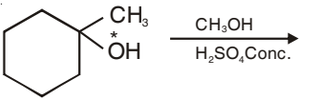
(2) 

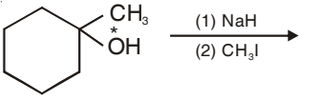
(3) 

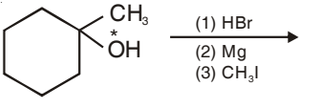
(4) 

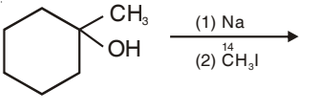
112. Match the Column (I) and (II).

Column I

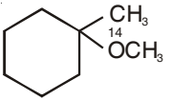
(1) 

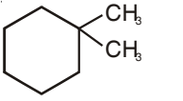
(2) 

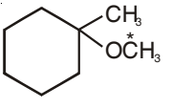
(3) 

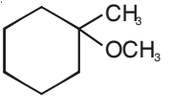
(4) 

Column II

(P) 

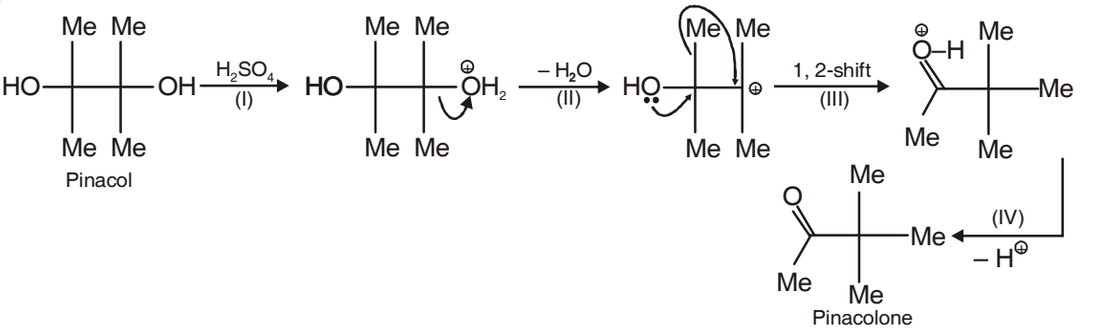
(Q) 

(R) 

(S) 

Paragraph for Q. Nos. 113 to 116

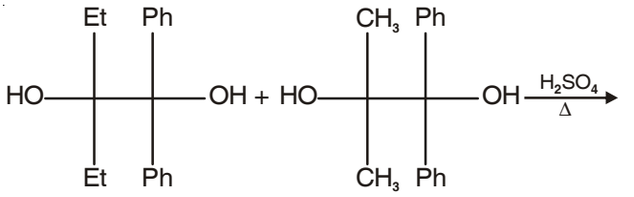
Di-tert-glycols rearrange in the presence of acid to give α-tertiary ketones. The trivial name of the simplest glycol of this type is pinacol, and this type of reaction therefore is named pinacol rearrangement (in this specific case, the reaction is called a pinacol-pinacolone rearrangement). The rearrangement involves 4 steps. One of the hydroxyl groups is protonated in the first step. A molecule of water is eliminated in the second step and a tertiary carbocation is formed. The carbocation rearranges in the third step inot a more stable carboxonium ion via a [1, 2] rearrangement. In the last step, the carboxonium ion is deprotonated and the product ketone is obtained.



113. What is R.D.S. of pinacol-pinacolone rearrangement?

(1) I Step (2) II Step

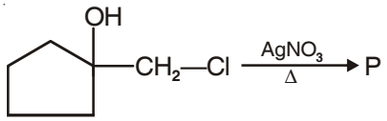
(3) III Step (4) IV Step

114. 

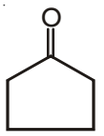
How many products obtained in above reaction?

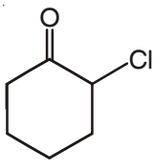
(1) 1 (2) 2

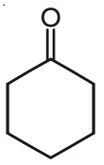
(3) 3 (4) 4

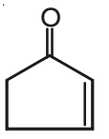
115. 

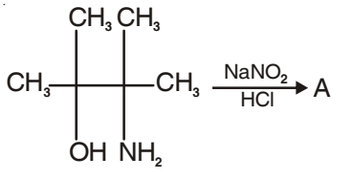
Product P is

(1) 

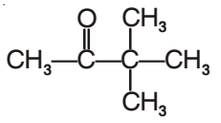
(2) 

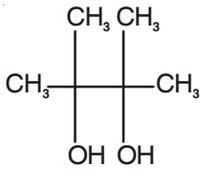
(3) 

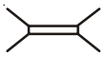
(4) 

116. 

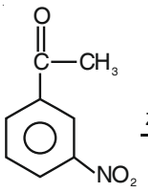
Product (A) is

(1) 

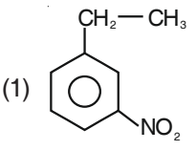
(2) 

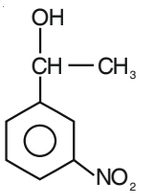
(3) 

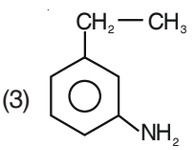
(4) None of these

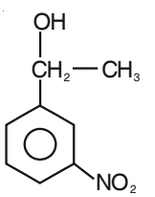
117. 

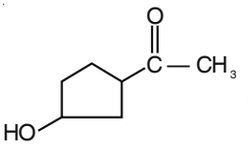
Zn(Hg), HCl ; Product of the Clemmensen reduction is

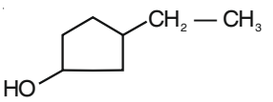
(1) 

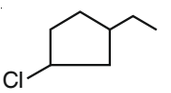
(2) 

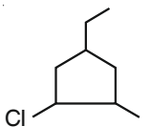
(3) 

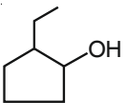
(4) 

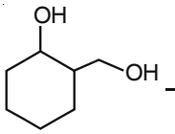
118.  $\xrightarrow[HCl]{Zn(Hg)}$ (A); Identify the A

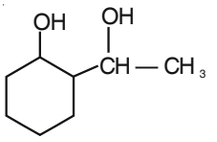
(1) 

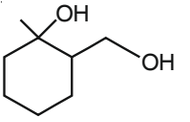
(2) 

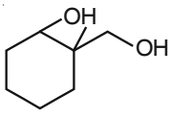
(3) 

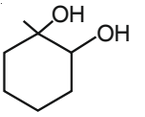
(4) 

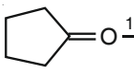
119.  $\xrightarrow{PCC\ (excess)}$ (A) $\xrightarrow[H^+]{1\ equivalent}$  (B) $\xrightarrow[H_3O^+]{CH_3MgBr}$ (C) $\xrightarrow{NaBH_4}$ (D). Product (D) will be

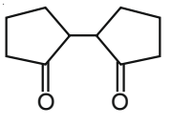
(1) 

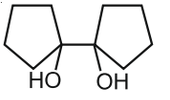
(2) 

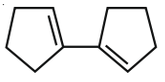
(3) 

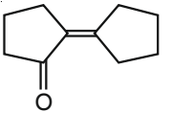
(4) 

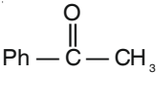
120. In the reaction,  $\xrightarrow[2.H_2O,\ heat]{1.Al(Hg),\ benzene,\ heat}$ X, the product (X) is

(1) 

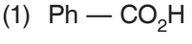
(2) 

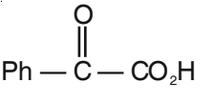
(3) 

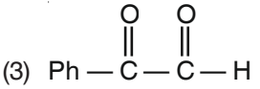
(4) 

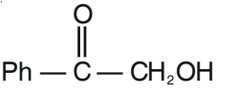
121.  $\xrightarrow[HCl]{NaNO_2}$ (A) $\xrightarrow[heat]{AC_2O}$ (B) $\xrightarrow{H_3O^+}$ (C)

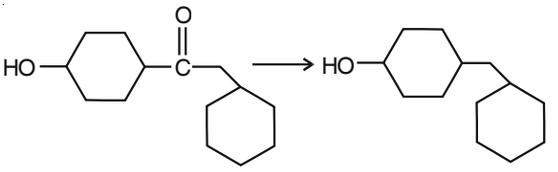
Product (C) of the above reaction is

(1) 

(2) 

(3) 

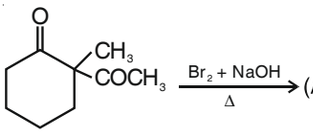
(4) 

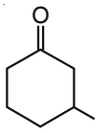
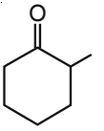
122. 

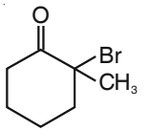
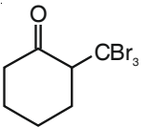
The above reduction can be best carried out by

(1) Clemmensen reduction (2) Wolff-Kishner reduction

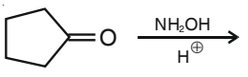
(3) $NaBH_4$ (4) None of these

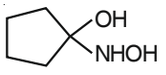
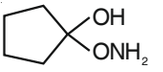
123.  $\xrightarrow[\Delta]{Br_2 + NaOH}$ (A) + $CHBr_3$; Product (C) of the reaction is $\xrightarrow[\Delta]{H^+}$ (C)

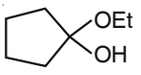
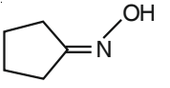
(1)  (2) 

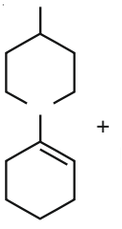
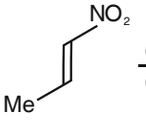
(3)  (4) 

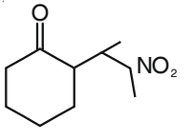
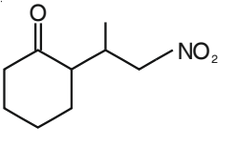
124. Which is the major product of the following reaction?

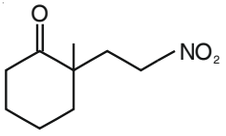
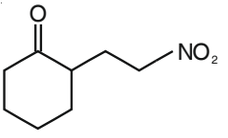


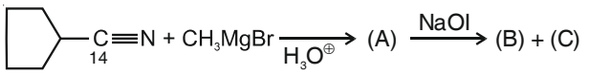
(1)  (2) 

(3)  (4) 

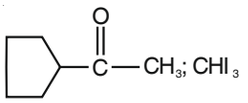
125.  +  $\xrightarrow[(2)\ H_3O^{\oplus}]{(1)\ Et_2O}$ (A); Product (A) of the reaction is 99%

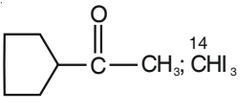
(1)  (2) 

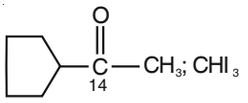
(3)  (4) 

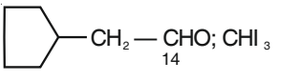
126. 

Product (A) and (C) is

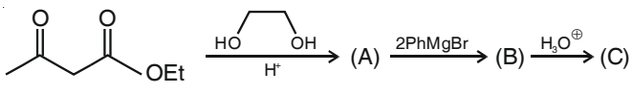
(1) 

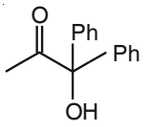
(2) 

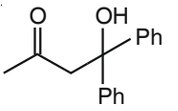
(3) 

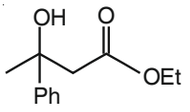
(4) 

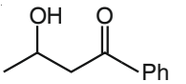
127. End product (C) of the reaction is



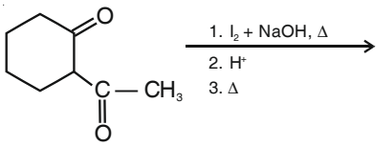
(1) 

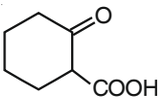
(2) 

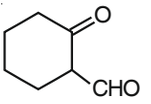
(3) 

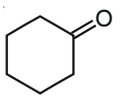
(4) 

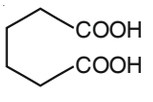
128. End products of the following sequence of reactions are

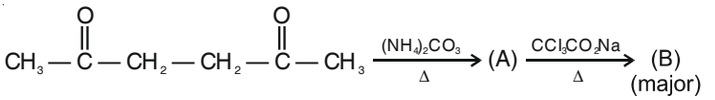


(1) Yellow ppt. of CHI_3, 

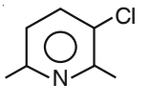
(2) Yellow ppt. of CHI_3, 

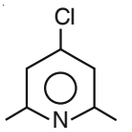
(3) Yellow ppt. of CHI_3, 

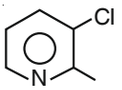
(4) Yellow ppt. of CHI_3, 

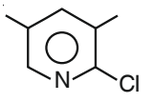
129. 

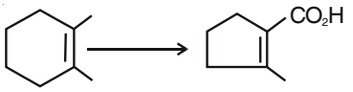
Product (B) of above reaction is

(1) 

(2) 

(3) 

(4) 

130. 

To carry out above conversion, arrange the following reagents in correct order.

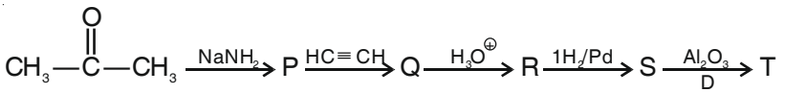
O_3/Zn (1) EtONa/EtOH/Δ (2) NaOCl (3) H^+ (4)

(1) $1 \rightarrow 3 \rightarrow 2 \rightarrow 4$

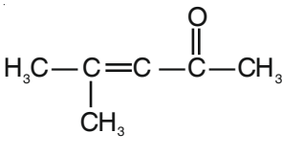
(2) $1 \rightarrow 2 \rightarrow 4 \rightarrow 3$

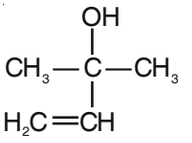
(3) $1 \rightarrow 3 \rightarrow 4 \rightarrow 2$

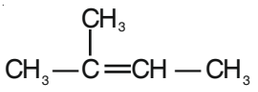
(4) $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$

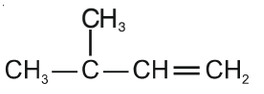
131. 

What is the final product T?

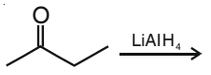
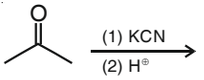
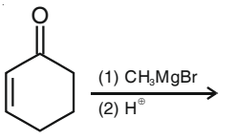
(1) 

(2) 

(3) 

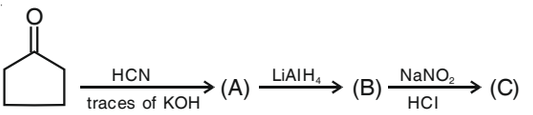
(4) 

132. Match the column

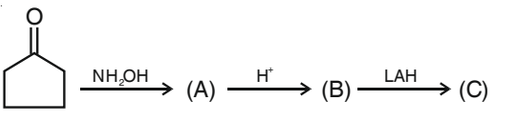
Column (I)	**Column (II)**
(1) 	(P) Racemic mixture
(2) 	(Q) Diastereomers
(3) $Ph - CH_2 - Cl \longrightarrow$	(R) Nu-addition reaction
(4) 	(S) Nu-Substitutions reaction

133. Match the column (I) and column (II).

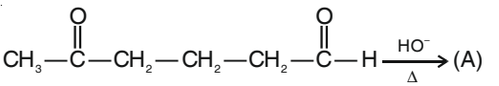
Column (I) **Column (II)**

(1) 

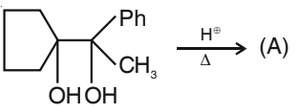
(P) Formation of six membering takes place

(2) 

(Q) Final product is ketone

(3) 

(R) Final product formed will give positive Tollens test

(4) 

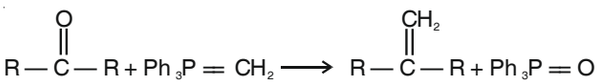
(S) Final product formed will react with 2, 4-DNP.

(2, 4-di-nitrophenyl hydrazine)

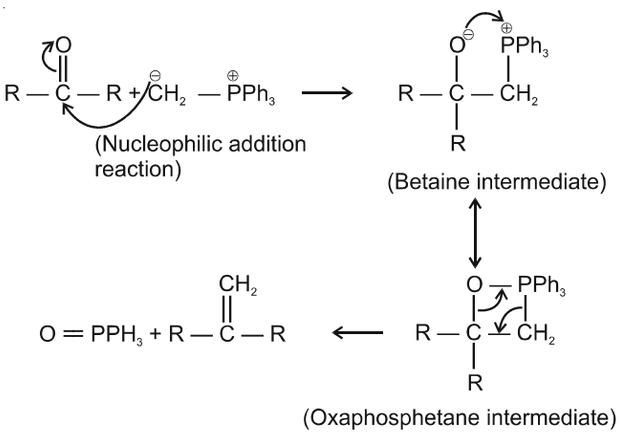
Paragraph for Q. Nos. 134 to 139

Wittig reaction:

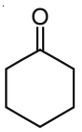
The reaction of a phosphorus ylide with an aldehyde (or) ketnoe introduces a carbon-carbon double bond is place of the carbonyl bond.

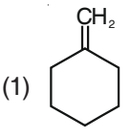


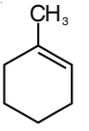
Mechanism:

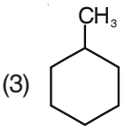


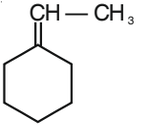
Driving force of the reaction is high bond energy of (P = O). (ΔH = –ve)

134.  + $PH_3P = CH_2 \longrightarrow$ (A) (major); Major product (A) is

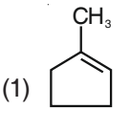
(1) 

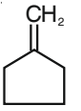
(2) 

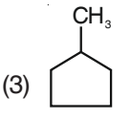
(3) 

(4) 

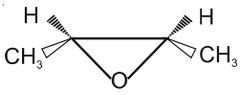
135.  (A); Major product (A) is

(1) 

(2) 

(3) 

(4) 

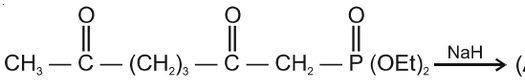
136.  (A); Major product (A) is

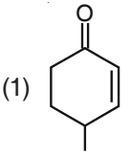
(1) cis-2-butene

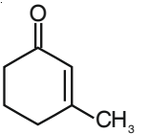
(2) trans-2-butene

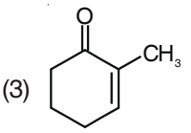
(3) iso-butene

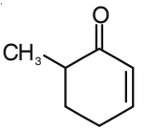
(4) 1-butene

137.  (A) (cyclic). Product (A) is

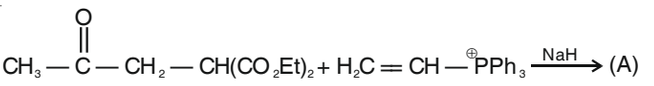
(1) 

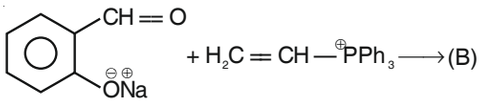
(2) 

(3) 

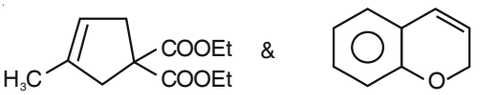
(4) 

138. Identify major product in given intramolecular Wittig reaction

Reaction-1 $\longrightarrow$ 

Reaction-2 $\longrightarrow$ 

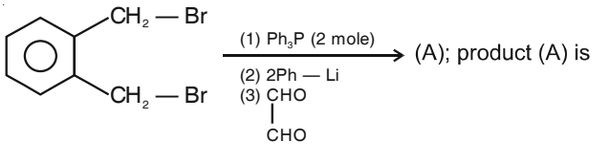
Product (A) and (B) respectively are

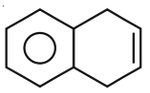
(1)  &

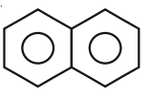
(2)  &

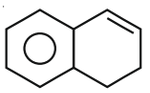
(3)  &

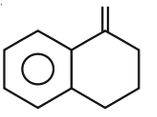
(4) 

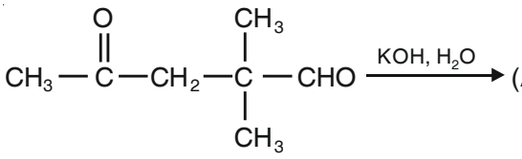
139.  (A); product (A) is

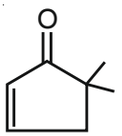
(1) 

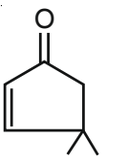
(2) 

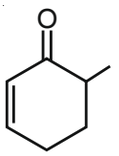
(3) 

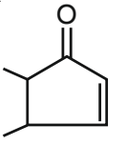
(4) 

140.  (A); Product A is

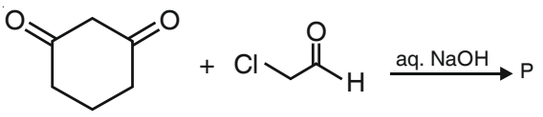
(1) 

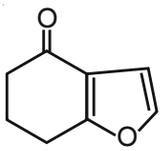
(2) 

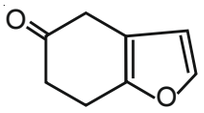
(3) 

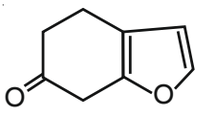
(4) 

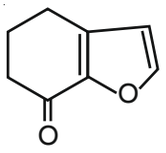
141. Identify the major product P in the following reaction:



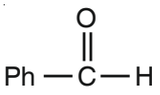
(1) 

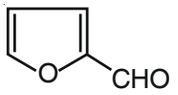
(2) 

(3) 

(4) 

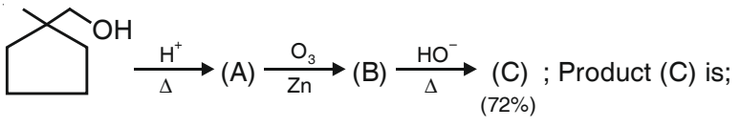
142. Which of the following compounds will not undergo Cannizzaro reaction, when treated with 50% aqueous alkali?

(1) 

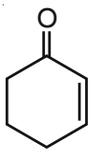
(2) 

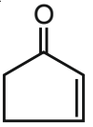
(3) Me_2CHCHO

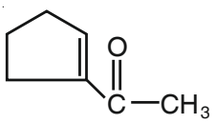
(4) $Ph-CH_2-CHO$

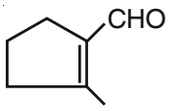
143. 

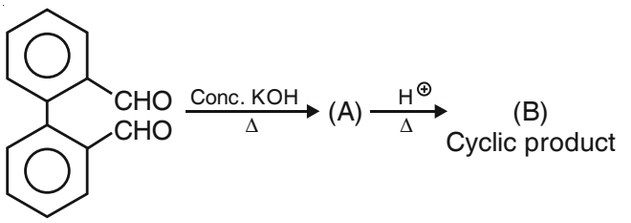
; Product (C) is;

(1) 

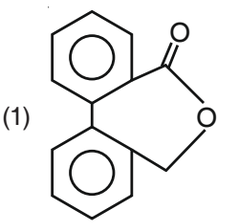
(2) 

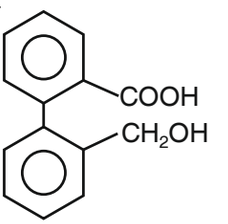
(3) 

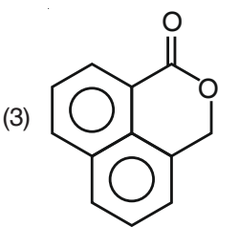
(4) 

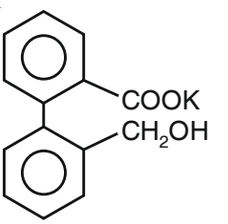
144. 

Structure of B is

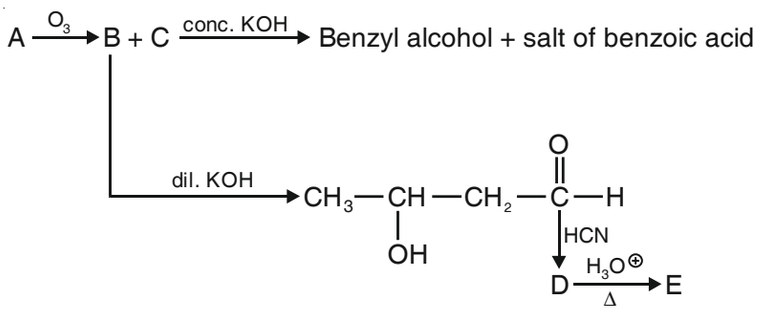
(1) 

(2) 

(3) 

(4) 

Paragraph for Q. Nos. 145 to 147



145. Structure of A is

(1) $H_2C=CH-CHO$

(2) $Ph-CH=CH-CH_3$

(3) $Ph-C(CH_3)=CH_2$

(4) $Ph-CH=C(CH_3)-CH_3$

146. Structure of (B) and (C) differentiated by

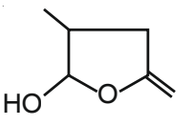
(1) Tollen's reagent

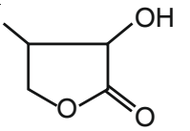
(2) Fehling solution

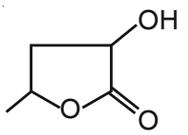
(3) 2, 4-DNP

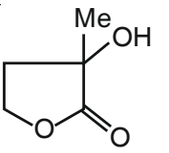
(4) $NaHSO_3$

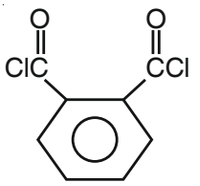
147. Structure of E is

(1) 

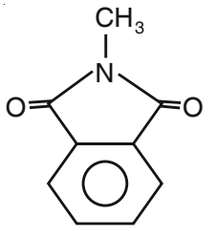
(2) 

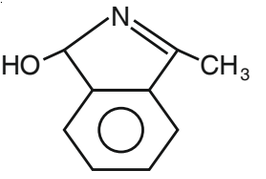
(3) 

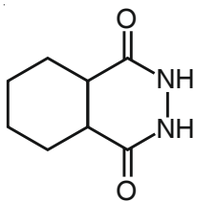
(4) 

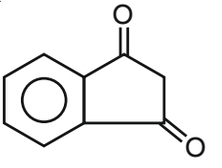
148. 
(A)

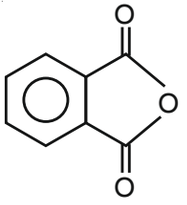
$+ CH_3 — NH_2 \longrightarrow$ Product of the reaction is :

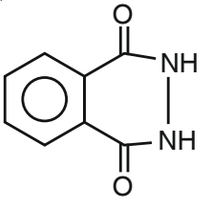
(1) 

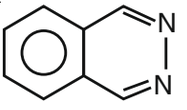
(2) 

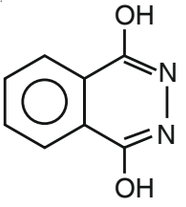
(3) 

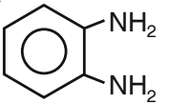
(4) 

149. 
$\xrightarrow{LiAlH_4}$ A $\xrightarrow{PCC}$ B $\xrightarrow{NH_2 — NH_2}$ C, Product (C) is :

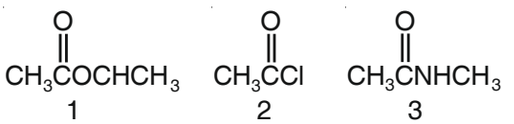
(1) 

(2) 

(3) 

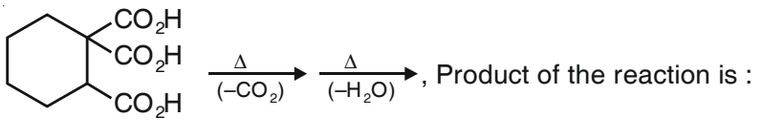
(4) 

150. Arrange the following in order of increasing reactivity (least → most) towards nucleophile



(1) 1 < 2 < 3 (2) 3 < 1 < 2

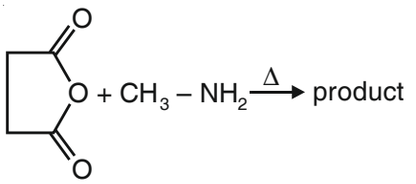
(3) 1 < 3 < 2 (4) 2 < 1 < 3

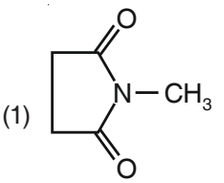
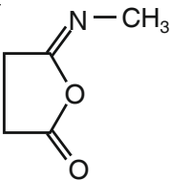
151.  , Product of the reaction is :

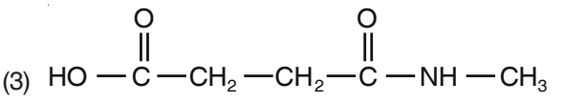
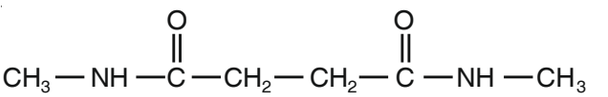
(1) cis-anhydride (2) trans-anhydride

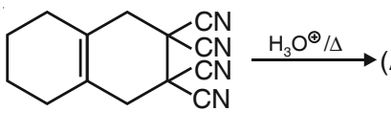
(3) Both (1) & (2) (4) Mono-basic acid

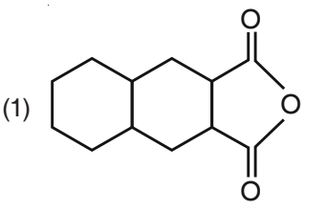
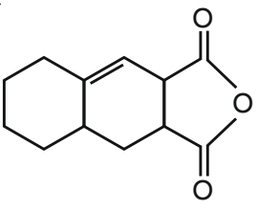
152. Which is the major product of the following reaction?

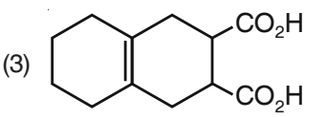
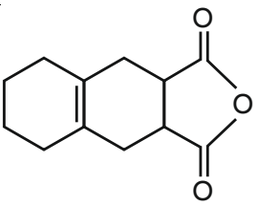


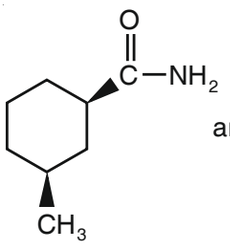
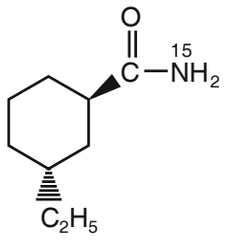
(1)  (2) 

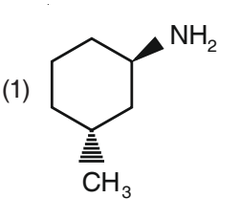
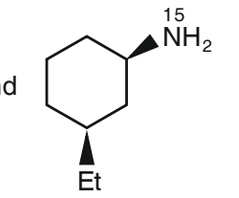
(3)  (4) 

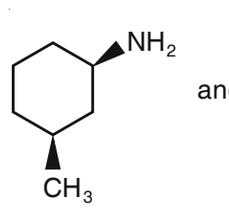
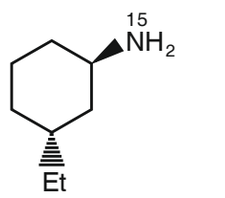
153.  (A); Product (A) of the reaction is:

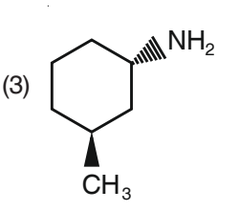
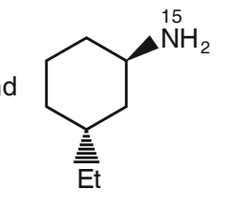
(1)  (2) 

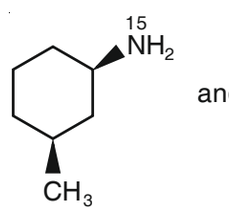
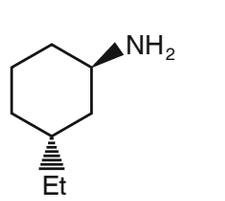
(3)  (4) 

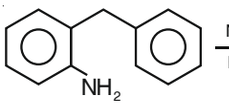
154.  and  $\xrightarrow[\text{(Intramolecular migration)}]{\overset{\ominus}{O}H/Br_2}$ Products are :

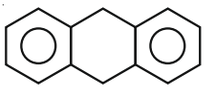
(1)  and 

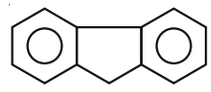
(2)  and 

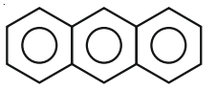
(3)  and 

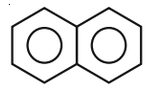
(4)  and 

155.  $\xrightarrow[H_2SO_4]{NaNO_2}$ (A); Product of this reaction is

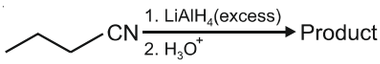
(1) 

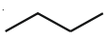
(2) 

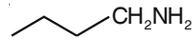
(3) 

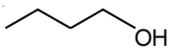
(4) 

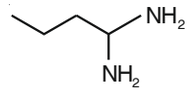
156. Choose the appropriate product for this reaction.



(1) 

(2) 

(3) 

(4) 

157. A nitrile X is treated with $LiAlH_4$ to obtain compound Y (C_2H_7N). In a separate reaction X is hydrolyzed in an acid medium to obtain Z. The product obtained after mixing Y and Z will be

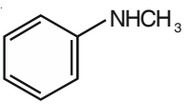
(1) $CH_3CONHCH_2CH_3$

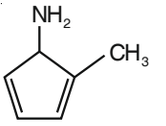
(2) $CH_3CH_2CONHCH_2CH_3$

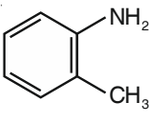
(3) $(CH_3COO^-)(CH_3CH_2NH_3^+)$

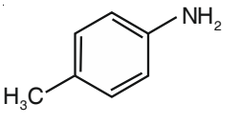
(4) $(CH_3CH_2COO^-)(CH_3NH_2^+)$

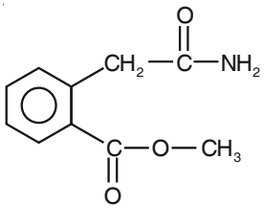
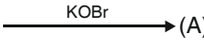
158. The compound X (C_7H_9N) reacts with benzensulfonyl chloride to give Y ($C_{13}H_{13}NO_2S$) which is insoluble in alkali. The compound X is

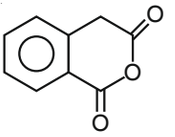
(1) 

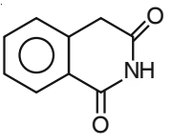
(2) 

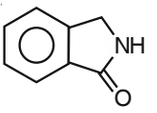
(3) 

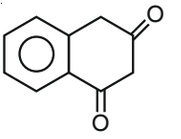
(4) 

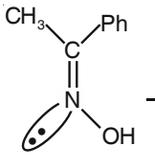
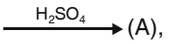
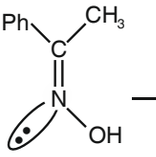
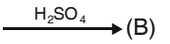
159.   (A); product (A) is

(1) 

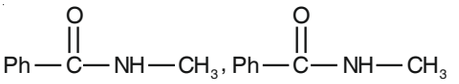
(2) 

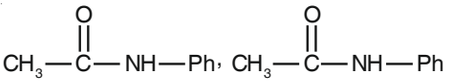
(3) 

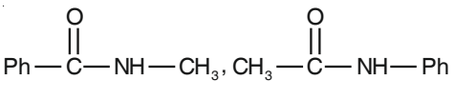
(4) 

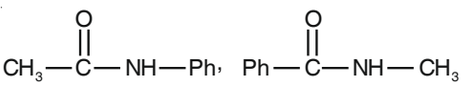
160.   (A),   (B)

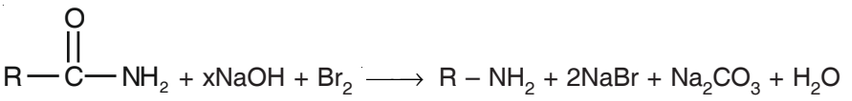
Product (A) and (B) respectively in the above reaction are

(1) 

(2) 

(3) 

(4) 

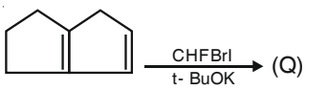
161. 

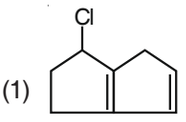
Number of moles of NaOH used in above Hoffmann bromamide reaction is

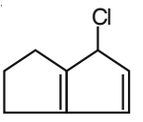
(1) 3 (2) 4

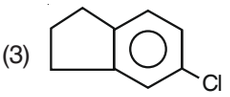
(3) 5 (4) 6

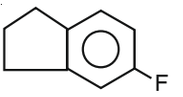
162. What is the product (Q) of the following reaction ?

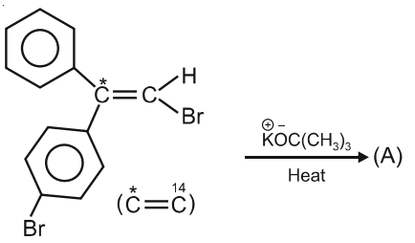


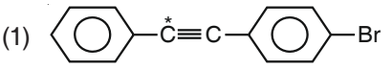
(1) 

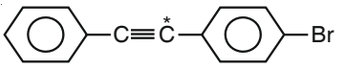
(2) 

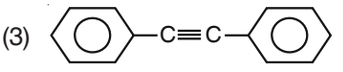
(3) 

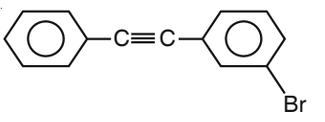
(4) 

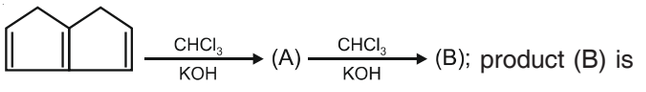
163. 

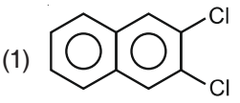
(1) 

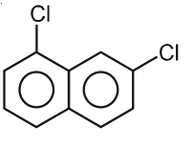
(2) 

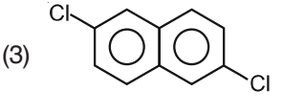
(3) 

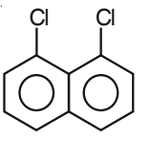
(4) 

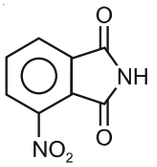
164. 

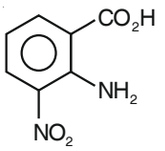
(1) 

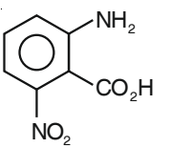
(2) 

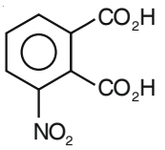
(3) 

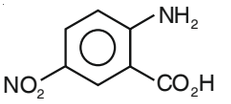
(4) 

165.  ; product X will be

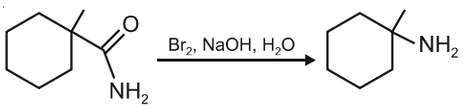
(1) 

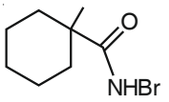
(2) 

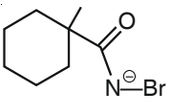
(3) 

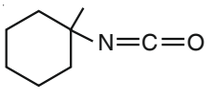
(4) 

166. Which of the following species would not be involved in the Hofmann rearrangement shown below?



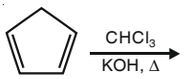
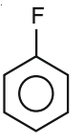
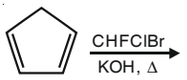
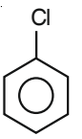
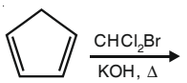
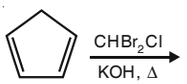
(1) 

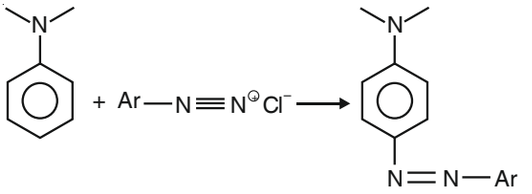
(2) 

(3) 

(4) All of the above are involved in the reaction

167. Match the Column (I) and (II).

Column I		**Column II**	
Reaction		Product	
(1)		(P)	
(2)		(Q)	
(3)		(R)	
(4)		(S)	

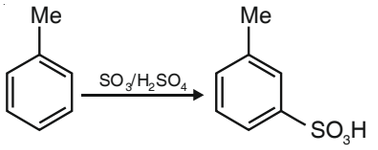
168. 

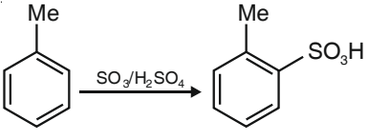
Above (C — N) coupling reaction take place at

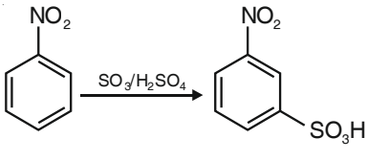
(1) Low pH (2) Intermediate pH

(3) High pH (4) Any pH

169. Rank the following reactions A, B and C in order of increasing rate,

(A) 

(B) 

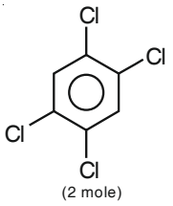
(C) 

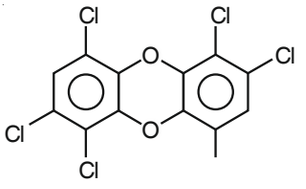
(1) (B) > (A) > (C)

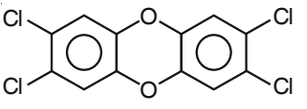
(2) (B) > (C) > (A)

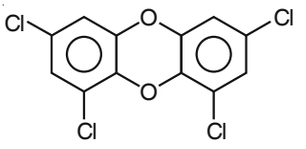
(3) (A) > (B) > (C)

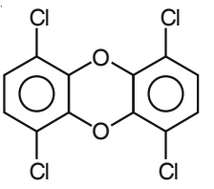
(4) (A) > (C) > (B)

170.  + $\underset{(4\text{ mole})}{HO^-} \longrightarrow \underset{(\text{major})}{(A)}$; product (A) of the given reaction is

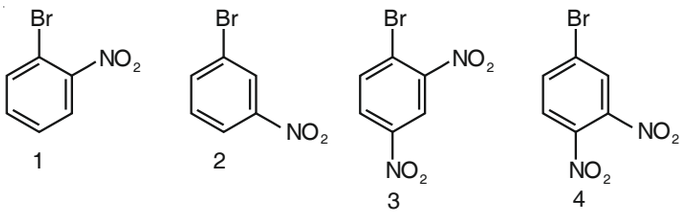
(1) 

(2) 

(3) 

(4) 

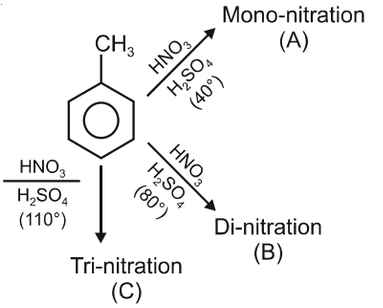
171. Rank the following in order of decreasing rate of reaction with alkoxide ion (CH_3CH_2O) in a nucleophilic aromatic substitution reaction :



(1) 3 > 4 > 1 > 2 (2) 3 > 4 > 2 > 1

(3) 2 > 1 > 4 > 3 (4) 4 > 3 > 2 > 1

172. How many products are capable of beings formed from toluene in each of following reaction?

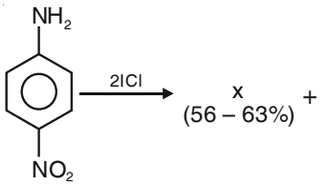


(1) A = 3, B = 6, C = 8

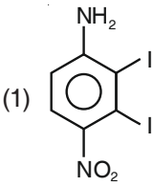
(2) A = 3, B = 6, C = 6

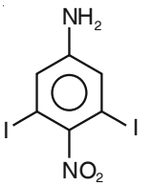
(3) A = 3, B = 6, C = 10

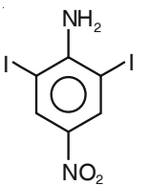
(4) A = 3, B = 4, C = 6

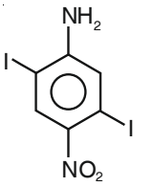
173. 

+ 2HCl; major product (x) in this reaction is

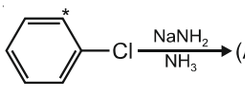
(1) 

(2) 

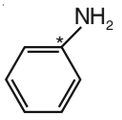
(3) 

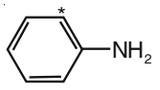
(4) 

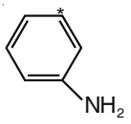
174. Two benzyne intermediates are likely to be formed equally. Reaction with amide ion can occur in two different directions with each benzyne, giving three possible products. They are formed in a 1 : 2 : 1 ratio. Asterisk (*) refers to ^{14}C.

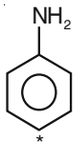


(A) product major; product (A) is

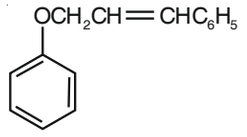
(1) 

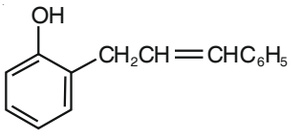
(2) 

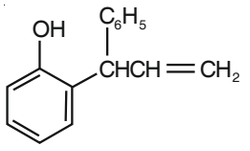
(3) 

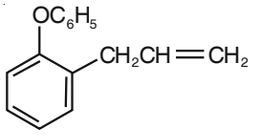
(4) 

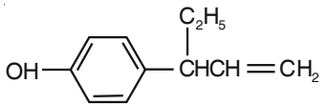
175. What is the product obtained by heating the following allylic ether of phenol?



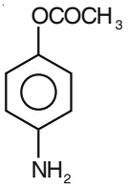
(1) 

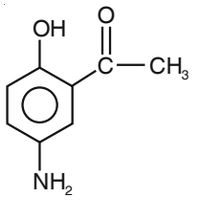
(2) 

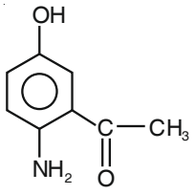
(3) 

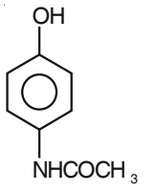
(4) 

176. *p*-aminophenol reacts with one equivalent of acetyl chloride in the presence of pyridine to give mainly

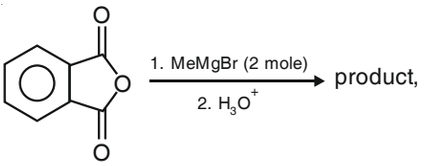
(1) 

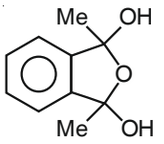
(2) 

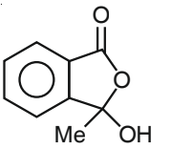
(3) 

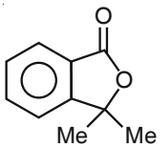
(4) 

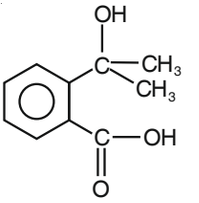
177. For the reaction, the product expected is :

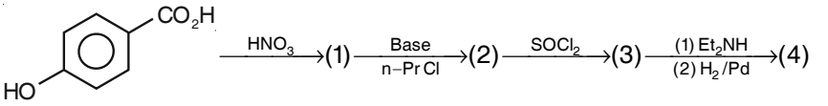


(1) 

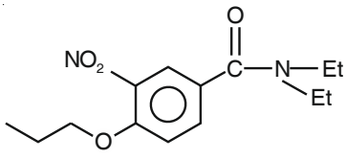
(2) 

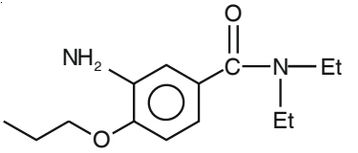
(3) 

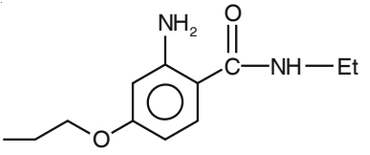
(4) 

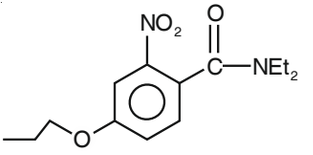
178. 

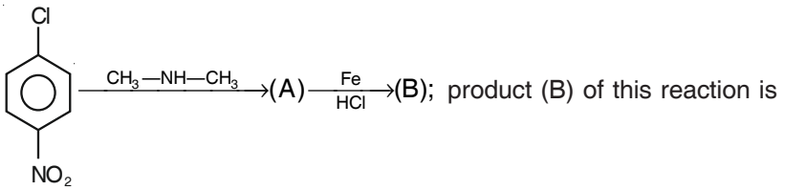
Product (4) in the above reaction is

(1) 

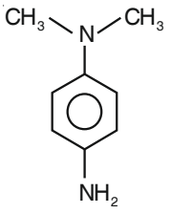
(2) 

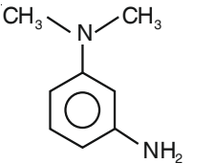
(3) 

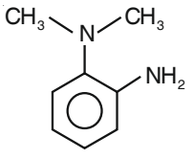
(4) 

179. 

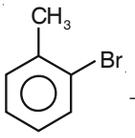
product (B) of this reaction is

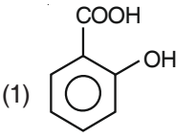
(1) 

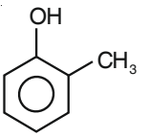
(2) 

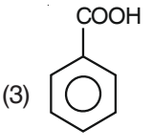
(3) 

(4) None of these

180.  $\xrightarrow[2.\ H_3O^+\\3.\ KMnO_4/OH^-\\4.\ H^+]{1.\ Mg/Ether}$ A, product A is

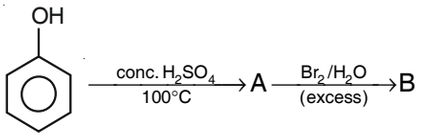
(1) 

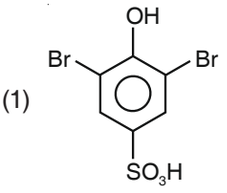
(2) 

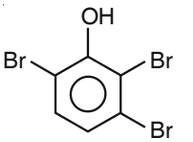
(3) 

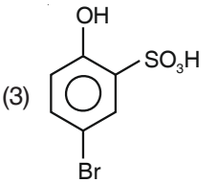
(4) 

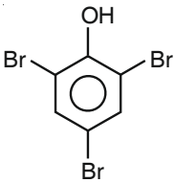
181. Identify the end product (B) of the following sequence of reactions.



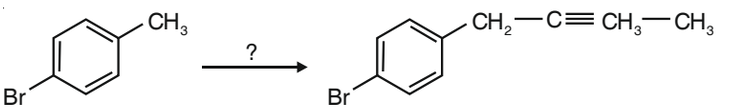
(1) 

(2) 

(3) 

(4) 

182. Which of the following procedures would be best for achieving the following reaction?



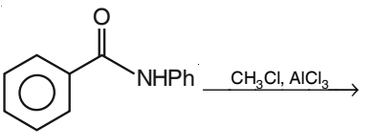
(1) (i) KOH and heat (ii) $CH_3C \equiv C - Br$

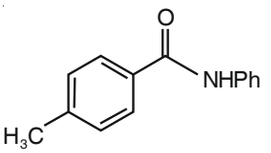
(2) (i) $KMnO_4$ and heat (ii) $CH_3C \equiv C^{(-)}Na^{(+)}$ (iii) excess H_2O

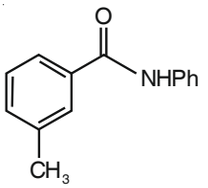
(3) (i) NBS in CCl_4 and heat (ii) $CH_3C \equiv C^{(-)}Na^{(+)}$

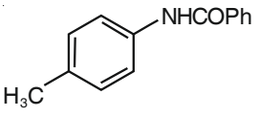
(4) (i) Mg in ether (ii) $CH_3C \equiv CBr$ (iii) excess H_3PO_4

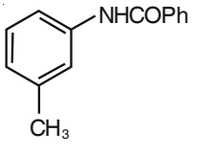
183. Give the major product of the following reactions :



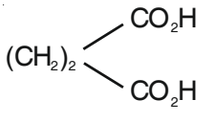
(1) 

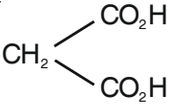
(2) 

(3) 

(4) 

184. Which of following acid give positive Tollen's reagent test?

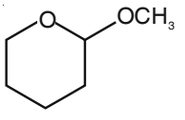
(1) 

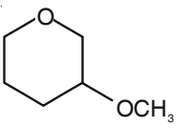
(2) 

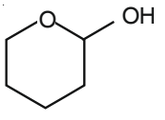
(3) 

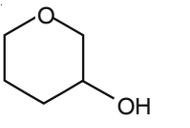
(4) HCO_2H

185. Which of the following compounds give positive Tollen's test?

(1) 

(2) 

(3) 

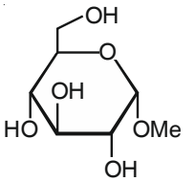
(4) 

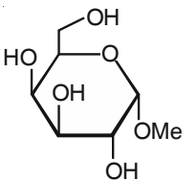
186. In Kjeldahl's method, nitrogen present is estimated as

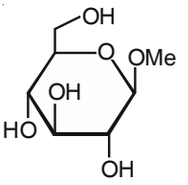
(1) N_2 (2) NH_3

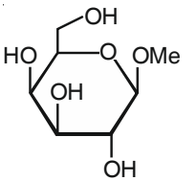
(3) NO_2 (4) None of these

187. Identify the correct set of stereochemical relationships amongst the following monosaccharides I - IV.

(I) 

(II) 

(III) 

(IV) 

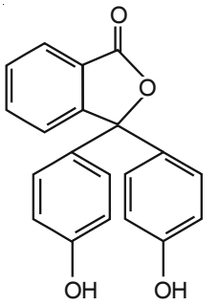
(1) I and II are anomers; III and IV are epimers

(2) I and II are epimers; III and IV are anomers

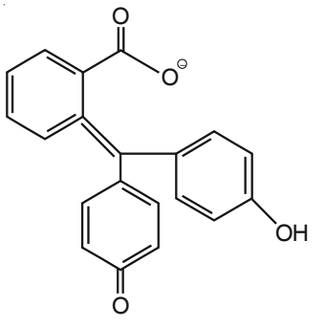
(3) I and III are anomers; I and II are epimers

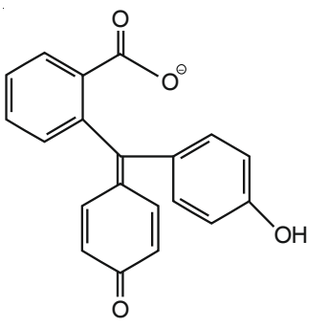
(4) I and III are epimers; II and IV are anomers

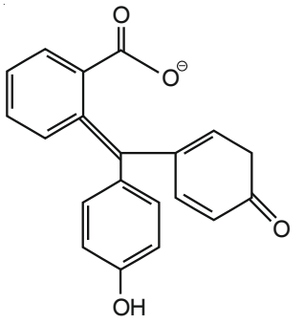
188. A dye, phenolphthalein is prepared by reacting phenol with phthalic anhydride in acidic medium. It gives pink colour in alkaline medium due to extended conjugation in a new complex formed (phthalein -dye test), identify the complex A


Phenolphthalein

$\xrightarrow{\ominus OH}$ (A) Product : The structure of (A) is

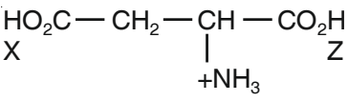
(1) 

(2) 

(3) 

(4) None of these

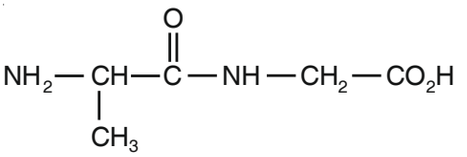
189. The pK_a values for the three ionizable groups X, Y and Z of glutamic acid are 4.3, 9.7 and 2.2 respectively



The isoelectric point for the amino acid is

(1) 7.00 (2) 3.25

(3) 4.95 (4) 5.95

190. 

Identify the amino acid obtained by hydrolysis of the above compound.

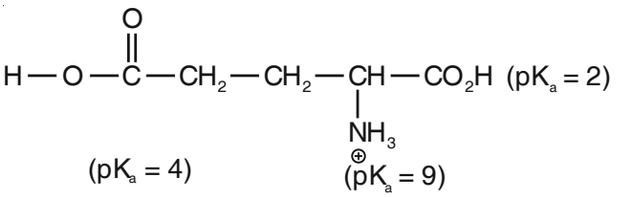
(1) Glycine

(2) Alanine

(3) Both (1) & (2)

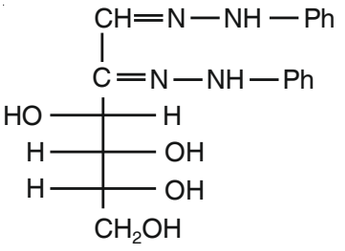
(4) None of these

191. Find iso-electric point of the given amino acid.



(1) 5.5 (2) 6.5

(3) 3 (4) 5

192. 

The given osazone can be obtained by

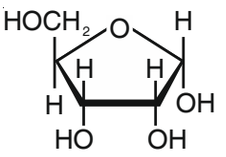
(1) D-Glucose

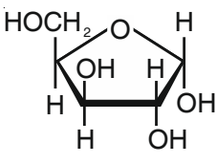
(2) D-Mannose

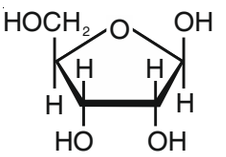
(3) D-Idose

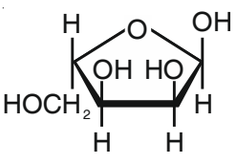
(4) Both (1) & (2)

193. Which of the following represents the anomer of the compound shown?



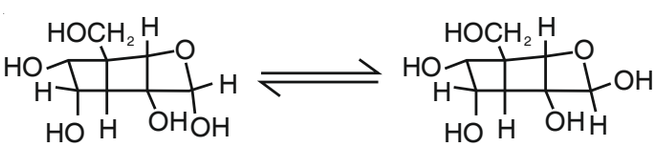
(1) 

(2) 

(3) 

(4) None of these

194. Which one of the statements concerning the equilibrium shown is true?



(1) The two structures are enantiomers of each other. They have equal but opposite optical rotations and recemize slowly at room temperature

(2) The two structures are enantiomers of each other. They racemize too rapidly at room temperature for their optical rotations to be measured

(3) The two structures are diastereomers of each other. Their interconversion is called mutarotation

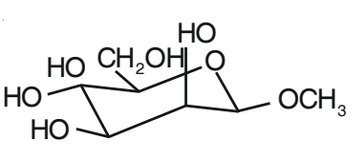
(4) The two structures are diastereomers of each other. Their interconversion does not require breaking and making bonds, only a change in conformation

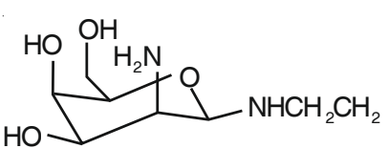
195. D-Glucose $\xrightarrow[\Delta]{HNO_3}$ (A); Product (A) is

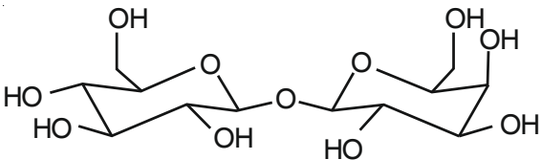
(1) D-Gluconic acid (2) D-Glucitol

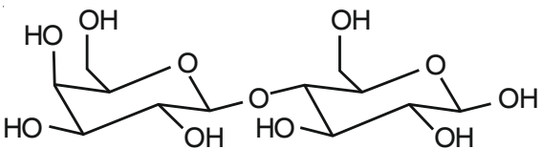
(3) D-Fructose (4) D-Glucaric acid

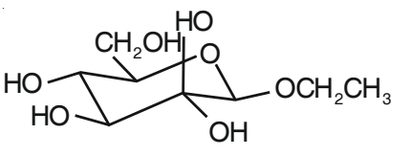
196. Which of the molecules below will react with Ag^+?

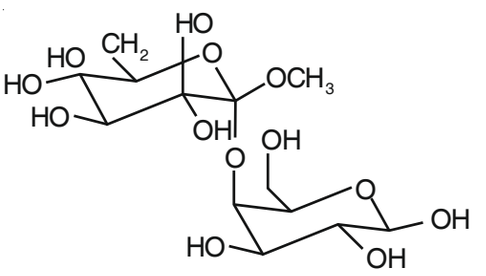
(i) 

(ii) 

(iii) 

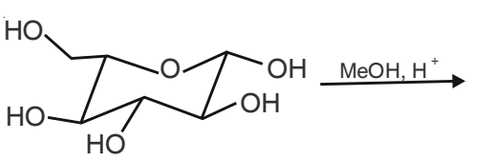
(iv) 

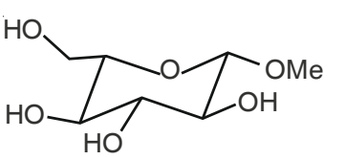
(v) 

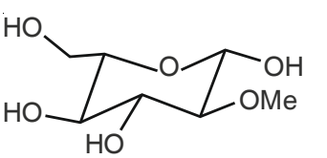
(vi) 

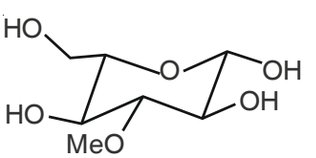
(1) (i), (iii) and (v)
(2) (ii) and (iv)
(3) (iv) and (vi)
(4) (i), (ii), (iii) and (vi)

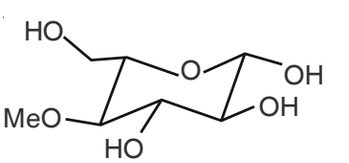
197. Predict the product of the following reaction



(1) 

(2) 

(3) 

(4) 

❑ ❑ ❑

Book Crunch

Mathematics

MATHEMATICS

1. $\lim_{n\to\infty}\left(\frac{n!}{(mn)^n}\right)^{1/n}$; $(m \in N)$ is equal to

   (A) $\frac{1}{em}$ (B) $\frac{m}{e}$

   (C) em (D) $\frac{e}{m}$

2. Let $f(x)=\left(\frac{e^{x\ln(2^x-1)}-(2^x-1)^x\sin x}{e^{x\ln x}}\right)^{\frac{1}{x}}$. Then right-hand limit of $f(x)$ at $x = 0$

   (A) Is equal to ln2 (B) Is equal to $\frac{\ln 2}{e}$

   (C) Is equal to e ln2 (D) Does not exist

3. If $f(x)=\sum_{\lambda=1}^{n}\left(x-\frac{1}{\lambda}\right)\left(x-\frac{1}{\lambda+1}\right)$, then $\lim_{n\to\infty} f(0)$ is

   (A) 1 (B) –1

   (C) 2 (D) 0

4. $\lim_{x\to 0^+}\left\{\lim_{n\to\infty}\left(\frac{[1^2(\sin x)^x]+[2^2(\sin x)^x]+\ldots+[n^2(\sin x)^x]}{n^3}\right)\right\}$, where [·] denotes the greatest integer function.

   (A) 0 (B) $\frac{2}{3}$

   (C) $\frac{4}{3}$ (D) $\frac{1}{3}$

5. Find $\lim_{x\to 0}\left\{\ln(1+x)-x+\frac{x^2}{2}-\frac{x^3}{3}+\ldots+(-1)^n\frac{x^n}{n}\right\}\times\frac{1}{x^{n+1}}$.

6. Evaluate $\lim_{n\to\infty}\frac{2^3-1}{2^3+1}\cdot\frac{3^3-1}{3^3+1}\cdot\frac{4^3-1}{4^3+1}\ldots\frac{n^3-1}{n^3+1}$.

7. Solve $\lim_{n\to\infty}\frac{1}{n^3}\{[[1^2x+1^2]]+[[2^2x+2^2]]+\ldots+[[n^2x+n^2]]\}$, where [·] denotes the greatest integer function.

8. Show that $\left(1+\sum_{k=1}^{n}\frac{2}{{}^nC_k}\right)^n \to e^2$ as $n \to \infty$ (for $n \geq 6$).

9. Suppose that function $f:\mathbb{R}\to\mathbb{R}$ satisfies the inequality $\left|\sum_{k=1}^{n}3^k\{f(x+ky)-f(x-ky)\}\right|\le 1$ for every positive integer n and for all $x, y\in\mathbb{R}$. Prove that f is a constant function.

10. If $n \in N$ and $a_n = 2^2 + 4^2 + 6^2 + ... + (2n)^2$ and $b_n = 1^2 + 3^2 + 5^2 + ... + (2n - 1)^2$. Find the value of $\lim_{n\to\infty}\frac{\sqrt{a_n}-\sqrt{b_n}}{\sqrt{n}}$.

11. At the end points A, B of a fixed segment of length L, lines are drawn meeting in C and making angles θ and 2θ respectively with the given segment. Let D be the foot of the altitude CD and let x represents the length of AD. Find the value of x as θ tends to zero, *i.e.*, $\lim_{\theta\to 0} x$.

12. At the end-points and the midpoint of a circular arc AB tangent lines are drawn, and the points A and B are joined. Prove that the ratio of the areas of the two triangles thus formed tends to 4 as the arc AB decreases indefinitely.

13. Evaluate the limit $\lim_{n\to\infty}\left(\frac{\sqrt{n^2+n}-1}{n}\right)^{2\sqrt{n^2+n}-1}$

14. Evaluate $\lim_{x\to\infty} x^2\sin\left(\ln\sqrt{\cos\frac{\pi}{x}}\right)$

15. Let $L=\prod_{n=3}^{\infty}\left(1-\frac{4}{n^2}\right)$; $M=\prod_{n=2}^{\infty}\left(\frac{n^3-1}{n^3+1}\right)$ and $N=\prod_{n=1}^{\infty}\frac{(1+n^{-1})^2}{1+2n^{-1}}$, then find the value of $L^{-1} + M^{-1} + N^{-1}$.

16. If $f(n,\theta)=\prod_{r=1}^{n}\left(1-\tan^2\frac{\theta}{2^r}\right)$, then compute $\lim_{n\to\infty} f(n,\theta)$.

17. Through a point A on a circle, a chord AP is drawn and on the tangent at A a point T is taken such that $AT = AP$. If TP produced meet the diameter through A at Q, prove that the limiting value of AQ, when P moves upto A is double the diameter of the circle.

Paragraph for Q. Nos. 18 to 20

Consider two functions $f(x)=\lim_{n\to\infty}\left(\cos\frac{x}{\sqrt{n}}\right)^n$ and $g(x) = -x^{4b}$, where $b=\lim_{x\to\infty}(\sqrt{x^2+x+1}-\sqrt{x^2+1})$. Then

18. $f(x)$ is

(A) e^{-x^2} (B) $e^{\frac{-x^2}{2}}$

(C) e^x (D) $e^{\frac{x^2}{2}}$

19. $g(x)$ is

(A) $-x^2$ (B) x^2

(C) x^4 (D) $-x^4$

20. Number of solutions of $f(x) + g(x) = 0$ is

(A) 2 (B) 4

(C) 0 (D) 1

21. Match Column-I with Column-II.

Column-I		Column-II	
A.	$\lim_{x\to\infty}\left(3-\sqrt[3]{x^3+x^2-1}+\sqrt[3]{x^3-x^2+1}\right)$ equals	(P)	$\frac{7}{3}$
B.	$\lim_{m\to\infty}\lim_{n\to\infty}\sum_{k=1}^{m}\frac{\sqrt[n]{(k-1)^n+k^n}}{m^2}$	(Q)	0
C.	$\lim_{n\to\infty}\left(\frac{n}{n^2-4}+\frac{(4)^n(-1)^n}{2^n-1}\right)^{-1}$ equals	(R)	$\frac{-5}{2}$
D.	$\lim_{x\to-\infty}\left\{x+\sqrt{x^2+5x\cos\frac{1}{\lvert x\rvert}}\right\}$ equals	(S)	$\frac{1}{2}$

22. What is the value of the $\left[\left\{4\sum_{r=1}^{n}(r+1)(r+2)(r+3)\right\}^{\frac{1}{4}}-n\right]$ is

(A) 0 (B) Does not exist

(C) $\frac{2}{5}$ (D) $\frac{5}{2}$

23. $\lim_{x\to\frac{\pi}{2}}\left\{\frac{\sqrt[3]{\sin^4 x+\cos^2 x+\sin^2 3x}}{\sin^4 x-\sin^4 2x-\sin^2 3x}-\frac{\sqrt[3]{\cos^2 x+\sin^4 2x+2\sin^2 3x}}{\sin^4 x-\sin^4 2x-\sin^2 3x}\right\}$ is equal to

(A) 2 (B) 0

(C) $\frac{1}{2\sqrt{2}}$ (D) $\frac{1}{3(2)^{2/3}}$

24. If $\lim_{x\to0}\frac{x^n\sin^n x}{x^n-\sin^n x}$ is zero, then the least positive integral value of n is

(A) 1 (B) 2

(C) 3 (D) 4

25. Let r^{th} term of a series be given by $t_r=\frac{r}{1-3r^2+r^4}$. Then $\lim_{n\to\infty}\sum_{r=1}^{n}t_r$ is

(A) $\frac{3}{2}$ (B) $\frac{1}{2}$

(C) $\frac{-1}{2}$ (D) $\frac{-3}{2}$

26. Let $f(x)$ be defined for all $x\in\mathbb{R}$ such that $\lim_{x\to0}\left[f(x)+\ln\left(1-\frac{1}{e^{f(x)}}\right)-\ln(f(x))\right]=0$; then $f(0)$ is equal to

(A) 0 (B) 1

(C) 2 (D) 3

27. The value of $\lim_{n\to\infty}\{(1.5)^n+([1+0.0001])^n\}^{1/n}$ is (where [ ] = G.I.F.)

(A) 1.5 (B) 2

(C) 3 (D) 2.5

28. If $f(x)=\begin{Bmatrix}2x-3; & -3\le x<0\\ 4-x^2; & 0\le x\le 5\end{Bmatrix}$ and $g(x)=\begin{Bmatrix}2x+3; & -10\le x<0\\ -1-x; & 0\le x\le 3\end{Bmatrix}$; then

(A) $\lim_{x\to 0^+} f(g(x))=-5$ (B) $\lim_{x\to 0^-} f(g(x))=-5$

(C) $\lim_{x\to 0} f(g(x))=-5$ (D) $\lim_{x\to 0} f(g(x))$ exists

Paragraph for Q. Nos. 29 to 31

Let $\lim_{n\to\infty}\left[\frac{2}{\pi}(1+n)\cos^{-1}\left(\frac{1}{n}\right)-n\right]=\lim_{n\to\infty} n\left(f\left(\frac{1}{n}\right)\right)$, then

29. $f'(0)$ equals

(A) $\frac{\pi-1}{\pi}$ (B) $\frac{\pi-2}{\pi}$

(C) $\pi-3$ (D) $\frac{\pi+2}{2\pi}$

30. $\lim_{x\to 0} f(x)$ equals

(A) 1 (B) 2

(C) 0 (D) $\frac{1}{2}$

31. $\left|\lim_{x\to\frac{1}{2}} f(x-1)\right|$ equals

(A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{2}}$

(C) $\frac{1}{3}$ (D) $\sqrt{2}$

32. If $\lim_{x\to\infty}\frac{x^3\sin\frac{1}{x}+x+1}{x^2+x+1}=\ell$, then evaluate $\sqrt{\ell\sqrt{\ell\sqrt{\ell\ldots\infty}}}$

33. $\lim_{n\to\infty}\frac{-3n+(-1)^n}{4n-(-1)^n}=\frac{r}{t}$ (lowest form), then evaluate $\sqrt{r^2+t^2}$.

34. If $\lim_{n\to\infty}\frac{\sqrt{n^3-2n^2+1}+\sqrt[3]{n^4+1}}{\sqrt[4]{n^6+6n^5+2}-\sqrt[5]{n^7+3n^3+1}}=k$, then evaluate $(k)^{2013}$.

35. If $\lim_{x\to\infty}\left(\sqrt{(x+a)(x+b)}-x\right)=f(a,b)$, then evaluate $f(9, 7)$.

36. Among the following, which is false?

(A) If f, g are any two continuous functions, then max. $\{f, g\}$; min. $\{f, g\}$ is also continuous

(B) If f is continuous and g is discontinuous, then $f + g$ is discontinuous

(C) If f is continuous and periodic in $[a, b]$, then it is bounded

(D) A periodic function which is discontinuous cannot have its period of discontinuity as period of the function

37. $f(x) = \frac{x^3}{4} - a\sin\pi x + 3;\ -4 \le x \le 4;\ a \in (0, 1)$. The value of $f(x)$ is $\frac{1999}{199}$ for some $x \in [-4, 4]$. This statement is

(A) True (B) False

(C) True only if $a \ge 0$ (D) True only if $a \in [-4, 4]$

38. If both $f(x)$ and $g(x)$ are differentiable functions at $x = x_0$, then the function defined as, $h(x)$ = maximum $\{f(x), g(x)\}$

(A) Is always differentiable at $x = x_0$ (B) Is never differentiable at $x = x_0$

(C) Is differentiable at $x = x_0$ if $f(x_0) \ne g(x_0)$ (D) Cannot be differentiable at $x = x_0$ if $f(x_0) = g(x_0)$

39. If $f(x)$ is twice differentiable function, then between two consecutive roots of the equation $f'(x) = 0$, then there exists

(A) More than one root of $f(x) = 0$ (B) At most one root of $f(x) = 0$

(C) Exactly one root of $f(x) = 0$ (D) At most one root of $f''(x) = 0$

40. Let $f(x) = x^3 - x^2 + x + 1$ and $g(x) = \begin{cases} \max\{f(t) \text{ for } 0 \le t \le x\}; & \text{for } 0 \le x \le 1 \\ 3 - x + x^2; & \text{for } 1 < x \le 2 \end{cases}$, then

(A) $g(x)$ is continuous and derivable at $x = 1$

(B) $g(x)$ is continuous but not derivable at $x = 1$

(C) $g(x)$ is neither continuous nor derivable at $x = 1$

(D) $g(x)$ is derivable but not continuous at $x = 1$

41. If $f(x) = \int_{-1}^{x} |t|\, dt,\ x \ge -1$, then

(A) f and f' are continuous for $x + 1 > 0$ (B) f is continuous but f' is not so for $x + 1 > 0$

(C) f and f' are not continuous at $x = 0$ (D) f is continuous at $x = 0$, but f' is not so

42. The function $f(x) = \frac{|x| - x(3^{1/x} + 1)}{3^{1/x} - 1};\ x \ne 0, f(0) = 0$ is

(A) Discontinuous at $x = 0$ (B) Continuous at $x = 0$ but not differentiable there

(C) Both continuous and differentiable at $x = 0$ (D) Differentiable but not continuous at $x = 0$

43. The function $f(x) = \begin{cases} 1, & |x| \ge 1 \\ \frac{1}{n^2}, & \frac{1}{n} < |x| < \frac{1}{n-1},\ n = 2, 3, \ldots \\ 0, & x = 0 \end{cases}$

(A) Is discontinuous at many points (B) Is continuous everywhere

(C) Is discontinuous at $x = \frac{1}{n},\ n \in \mathbb{Z} - \{0\}$ (D) $f(x)$ is continuous at $x = 0$

44. Let $f(x)=\left[\frac{1}{x}[x]\right]; x\neq 0, f(0)=0,$ then ([ ] is the greatest integer function)

(A) f is not continuous at $x = 0$ (B) f is continuous at $x = 0$

(C) f is discontinuous at $x = 1$ (D) f is continuous at $x = 1$

45. Let $f(x) = \cos x$ and $H(x)=\begin{cases}\min.\{f(t):0\le t\le x\}; & \text{for } 0\le x\le \frac{\pi}{2}\\ \frac{\pi}{2}-x; & \text{for } \frac{\pi}{2}<x\le 3\end{cases}$, then

(A) $H(x)$ is continuous and derivable in [0, 3]

(B) $H(x)$ is continuous but not derivable at $x=\frac{\pi}{2}$

(C) $H(x)$ is neither continuous nor derivable at $x=\frac{\pi}{2}$

(D) Maximum value of $H(x)$ in [0, 3] is 1

46. If $f(x)=\begin{cases}x^n\cdot\frac{e^{1/x}}{1+e^{1/x}}; & x\neq 0\\ 0; & x=0\end{cases}$; then

(A) If $n = 1$, function is continuous and differentiable

(B) If $n = 2$, function is continuous and differentiable

(C) If $n = 0$, function is discontinuous and non-differentiable

(D) If $n = 0$, function is discontinuous and differentiable

Paragraph for Q. Nos. 47 to 49

Left hand derivative and right hand derivative of a function $f(x)$ at a point $x = a$ are defined as

$$f'(a^-)=\lim_{h\to 0^+}\frac{f(a)-f(a-h)}{h}=\lim_{h\to 0^-}\frac{f(a+h)-f(a)}{h}=\lim_{x\to a^-}\frac{f(x)-f(a)}{x-a}$$

and $$f'(a^+)=\lim_{h\to 0^+}\frac{f(a+h)-f(a)}{h}=\lim_{h\to 0^-}\frac{f(a)-f(a-h)}{h}=\lim_{x\to a^+}\frac{f(a)-f(x)}{a-x}$$

respectively. Let f be a twice differentiable function. Then answer the following questions.

47. If f is odd, which of the following is right hand derivative of f at $x = -a$?

(A) $\lim_{h\to 0^-}\frac{f(a-h)-f(a)}{-h}$ (B) $\lim_{h\to 0^-}\frac{f(h-a)-f(a)}{h}$

(C) $\lim_{h\to 0^+}\frac{f(a)+f(a-h)}{-h}$ (D) $\lim_{h\to 0^-}\frac{f(-a)-f(-a-h)}{-h}$

48. If f is even, which of the following is right hand derivative of f' at $x = a$?

(A) $\lim_{h\to 0^-}\frac{f'(a)+f'(-a+h)}{h}$ (B) $\lim_{h\to 0^+}\frac{f'(a)+f'(-a-h)}{h}$

(C) $\lim_{h\to 0^+}\frac{-f'(-a)+f'(-a-h)}{-h}$ (D) $\lim_{h\to 0^+}\frac{f'(a)+f'(-a+h)}{-h}$

49. If $\lim_{h\to 0}\frac{f(-x)-f(-x-h)}{h}=\lim_{h\to 0}\frac{f(x)-f(x-h)}{-h}$, where h is real, then

(A) f is odd (B) f is even

(C) f is neither odd nor even (D) Nothing can be concluded

Paragraph for Q. Nos. 50 to 55

Let us define $f:[-\pi,\pi]\to\mathbb{R}$, $g:[-\pi,\pi]\to\mathbb{R}$ by

$f(x)$ = min. {sinx, cos2x}

$g(x)$ = min. {sinx, cos2x}

Answer the following questions.

50. Number of points where $f(x)$ is continuous but not differentiable is

(A) 2 (B) 3

(C) 1 (D) 0

51. Number of points where $|g(x)|$ is continuous but not differentiable is

(A) 2 (B) 3

(C) 4 (D) 5

52. Range of $f(x)$ is

(A) $\left[\frac{-1}{2},1\right]$ (B) $\left[-1,\frac{1}{2}\right]$

(C) [–1, 1] (D) $\left[\frac{1}{2},1\right]$

53. Range of $g(x)$ is

(A) $\left[\frac{-1}{2},1\right]$ (B) $\left[-1,\frac{1}{2}\right]$

(C) [–1, 1] (D) $\left[\frac{1}{2},1\right]$

54. $f(x)$ is increasing in the interval

(A) $\left[-\pi,\frac{-\pi}{2}\right]$ (B) $\left[\frac{-\pi}{2},\frac{\pi}{6}\right]$

(C) $\left[\frac{\pi}{2},\frac{5\pi}{6}\right]$ (D) $\left[\frac{-\pi}{2},\frac{\pi}{2}\right]$

55. $g(x)$ is decreasing in the interval

(A) $\left[-\pi,\frac{-\pi}{2}\right]$ (B) $\left[0,\frac{\pi}{6}\right]$

(C) $\left[\frac{\pi}{2},\frac{5\pi}{6}\right]$ (D) $\left[-\pi,\frac{\pi}{2}\right]$

Paragraph for Q. Nos. 56 to 58

A function $f(x)$ having the following properties :

(i) $f(x)$ is continuous except at $x = 3$

(ii) $f(x)$ is differentiable except at $x = -2$ and $x = 3$

(iii) $f(0) = 0, \lim_{x\to 3} f(x) \to -\infty, \lim_{x\to -\infty} f(x) = 3, \lim_{x\to\infty} f(x) = 0$

(iv) $f'(x) > 0 \ \forall \ x \in (-\infty, 2) \cup (3, \infty)$ and $f'(x) \le 0 \ \forall \ x \in (-2, 3)$

$f''(x) > 0 \ \forall \ x \in (-\infty, -2) \cup (-2, 0)$ and $f''(x) < 0 \ \forall \ x \in (0, 3) \cup (3, \infty)$

Then answer the following questions.

56. Maximum possible number of solutions of $f(x) = |x|$ is

(A) 2 (B) 1

(C) 3 (D) 4

57. Graph of function $y = f(-|x|)$ is

(A) Differentiable for all x, if $f'(0) = 0$

(B) Continuous but not differentiable at two points, if $f'(0) = 0$

(C) Continuous but not differentiable at one points, if $f'(0) = 0$

(D) Discontinuous at two points if $f'(0) = 0$

58. $f(x) + 3x = 0$ has five solutions, if

(A) $f(-2) > 6$ (B) $f'(0) < -3$ and $f(-2) > 6$

(C) $f'(0) > -3$ (D) $f'(0) > -3$ and $f(-2) > 6$

59. Match Column-I with Column-II.

	Column-I	**Column-II**
A.	The function $f(x) = \|x^2 + (\lambda - 1)\|x\| - \lambda\|$ is non-differentiable at five points for λ equals	(P) -1
B.	The function defined by $f(x) = \begin{cases} \min.(\|x\|, \sqrt{1-x^2}); & -1 \le x \le 1 \\ (\|x\|); & 1 < \|x\| < 2 \end{cases}$ is non-differentiable at	(Q) $\frac{-1}{\sqrt{2}}$
C.	$f(x) = \|x + 1\|(\|x\| + \|x - 1\|)$ is non-differentiable in interval $[-2, 2]$ at the points x equals	(R) $\frac{1}{\sqrt{2}}$
D.	If $f\left(\frac{x+y}{2}\right) = \frac{f(x)+f(y)}{2} \ \forall \ x, y \in \mathbb{R}$. If $f'(0) = -1$ and $f(0) = 1$, then $f(2)$ is less than or equal to	(S) 1
		(T) 0

60. Let $g(x) = \frac{(x-1)^n}{\ln \cos^m (x-1)}; \ 0 < x < 2$, m and n are integers, $m \ne 0$, $n > 0$ and let p be the left hand derivative of $|x - 1|$ at $x = 1$. If $\lim_{x\to 1^+} g(x) = p$, then evaluate $(m)^n$

61. Let [x] denotes the greatest integer function and $f(x)$ be defined as $f(x)=\begin{cases}\dfrac{(\exp\{(x+2)\ln 4\})^{\frac{[x+1]}{4}}-16}{4^x-16}; & x<2\\ a\cdot\dfrac{1-\cos(x-2)}{(x-2)\tan(x-2)}; & x>2\end{cases}$.

Find the value of a for which $f(x)$ may be continuous at $x = 2$.

62. If $f(x)=\begin{cases}|4x-5|\,[x]; & \text{for } x>1\\ [\cos\pi x]; & \text{for } x\le 1\end{cases}$; where [·] is greatest integer function, then m is the number of points of discontinuity of $f(x)$ and n is the number of points of non-differentiability in [0, 2], then evaluate $(m + n)$.

63. A function $f:\mathbb{R}\to\mathbb{R}$ is defined as $f(x)=\lim\limits_{n\to\infty}\dfrac{ax^2+bx+c+e^{nx}}{1+ce^{nx}}$, where $f(x)$ is continuous on $\mathbb{R}$. Then find the value of c.

64. Let $f(x)$ is a differentiable function on [7, 10] and takes only irrational values, given $f(8)=\sqrt{3}$, then evaluate $\prod\limits_{k=7}^{10} f(k)$.

65. Find the number of points where $f(x)=\dfrac{1}{\ln[x^2-3x+3]}$ is discontinuous.

66. If $\dfrac{\cos^4\theta}{x}+\dfrac{\sin^4\theta}{y}=\dfrac{1}{x+y}$, then $\dfrac{dy}{dx}$ =

(A) xy (B) $\tan^2\theta$

(C) 0 (D) $(x^2 + y^2)\sec^2\theta$

67. If $f(x)=e^x-e^{-x}-2\sin x-\dfrac{2}{3}x^3$, then the least value of n for which $\dfrac{d^n}{dx^n}f(x)$ at $x = 0$ is non-zero is

(A) 2 (B) 1

(C) 7 (D) Either 1 or 2

68. If $\prod\limits_{r=1}^{n}\cos\left(\dfrac{x}{2^r}\right)=\dfrac{\sin x}{2^n\sin\left(\dfrac{x}{2^n}\right)}$, then which of the following statement(s) is/are correct?

(A) $\sum\limits_{r=1}^{n}\dfrac{1}{2^r}\tan\left(\dfrac{x}{2^r}\right)=\dfrac{1}{2^n}\cot\left(\dfrac{x}{2^n}\right)-\cot x$

(B) $\sum\limits_{r=1}^{n}\dfrac{1}{2^r}\tan\left(\dfrac{x}{2^r}\right)=\dfrac{1}{2^n}\cot\left(\dfrac{x}{2^n}\right)$

(C) $\sum\limits_{r=1}^{n}\dfrac{1}{2^{2r}}\sec^2\left(\dfrac{x}{2^r}\right)=\dfrac{1}{2^n}\operatorname{cosec}^2\left(\dfrac{x}{2^n}\right)-\operatorname{cosec}^2 x$

(D) $\sum\limits_{r=1}^{n}\dfrac{1}{2^{2r}}\sec^2\left(\dfrac{x}{2^r}\right)=\operatorname{cosec}^2 x-\dfrac{1}{2^{2n}}\operatorname{cosec}^2\left(\dfrac{x}{2^n}\right)$

69. If $y = \cos^{-1}\frac{x^n - x^{-n}}{x^n + x^{-n}}$, then $y'(x)$ is equal to

(A) $\frac{2nx^{n-1}}{x^{2n}+1}$ if n is even

(B) $\frac{2nx^n}{|x|(x^{2n}+1)}$ if n is odd

(C) $-\frac{2nx^n}{|x|(x^{2n}+1)}$ if n is odd

(D) $\frac{2nx^{n-1}}{(x^{2n}+1)}$

70. If $y = \int_0^x f(t)\sin\{p(x-t)\}\,dt$, $\frac{d^2y}{dx^2} + p^2y$ equals

(A) 0

(B) y

(C) $k \cdot f(x)$

(D) $k^2 f(x)$

71. If $x < 1$, then the value of $\frac{1-2x}{1-x+x^2} + \frac{2x-4x^3}{1-x^2+x^4} + \frac{4x^3-8x^7}{1-x^4+x^8} + \ldots \infty$ is equal to

(A) $\frac{1}{1+x+x^2}$

(B) $\frac{1+2x}{1+x+x^2}$

(C) $\frac{1-x+x^2}{1+x+x^2}$

(D) 1

72. If $x^2e^y + 2xye^x + 13 = 0$, then $\frac{dy}{dx}$ =

(A) $-\frac{2xe^{y-x}+2y(x+1)}{x(xe^{y-x}+2)}$

(B) $\frac{2xe^{x-y}+2y(x+1)}{x(xe^{y-x}+2)}$

(C) $-\frac{2xe^{x-y}+2y(x+1)}{x(xe^{x-y}+2)}$

(D) $\frac{2xe^{x-y}+2x(y+1)}{x(xe^{x-y}+2)}$

73. If $f(x) = \frac{g(x)+g(-x)}{2} + \frac{2}{[h(x)+h(-x)]^{-1}}$, where g and h are differentiable function, then $f'(0)$ is

(A) 1

(B) $\frac{1}{2}$

(C) $\frac{3}{2}$

(D) 0

74. If $y = Ae^{x^2} + Be^{x^2/2}$ satisfies the relation $\frac{d^2y}{dx^2} + \left(kx - \frac{1}{x}\right)\frac{dy}{dx} + 2x^2y = 0$, then the value of k is equal to

(A) 2

(B) 1

(C) –3

(D) 3

75. If $y = \frac{x^2}{2} + \frac{1}{2}x\sqrt{x^2+1} + \ln\sqrt{x+\sqrt{x^2+1}}$ satisfies the relation $ky = x\frac{dy}{dx} + \ln\left(\frac{dy}{dx}\right)$, then the value of k is equal to

(A) 2

(B) 1

(C) 3

(D) 0

76. Let $f(t) = \log_3 t$. Then $\frac{d}{dx}\left\{\int_{x^2}^{x^3} f(t)\,dt\right\}$

(A) Has a value 0 when $x = 0$

(B) Has a value of 0 when $x = 1$, $x = \frac{4}{9}$

(C) Has a value $\frac{9e^2 - 4e}{\ln 3}$ when $x = e$

(D) Has a differential coefficient $\frac{(27e - 8)}{\ln 3}$ when $x = e$

77. Match Column-I with Column-II.

Column-I	**Column-II**
A. The function y defined by the equation $xy - \log y = 1$ satisfies $x(yy'' + y'^2) - y'' + kyy' = 0$. The value of k is	(P) 24
B. If the function $y(x)$ represented by $x = \sin t$, $y = ae^{r\sqrt{2}} + be^{r\sqrt{2}}$, $t \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ satisfies the equation $(1 - x^2)y'' - xy' = ky$, then k is equal to	(Q) 2
C. Let $F(x) = f(x)g(x)h(x)$ for all real x, where $f(x)$, $g(x)$ and $h(x)$ are differentiable functions. At some point x_0, if $F'(x_0) = 21\ F(x_0)$, $f'(x_0) = 4\ f(x_0)$, $g'(x_0) = -7g(x_0)$ and $h'(x_0) = kh(x_0)$, then k is equal to	(R) 4
D. Let $f(x) = x^n$, n being a non-negative integer. The number of values of n for which the equality $f'(a + b) = f'(a) + f'(b)$ is valid for all, a, $b > 0$, is	(S) 3

78. Tangent to the folium of descartes $x^3 + y^3 = 3axy$ at the point where it meets the parabola $y^2 = ax$ are parallel to

(A) x-axis

(B) y-axis

(C) $y = x$

(D) $y = -x$

79. The tangent and normal at the point $P(at^2, 2at)$ to the parabola $y^2 = 4ax$ meet the x-axis in T and N respectively, then the angle at which the tangent at P to the parabola is inclined to the tangent at P to the circle through T, P, N is

(A) $\tan^{-1}t^2$

(B) $\cot^{-1}t^2$

(C) $\tan^{-1}|t|$

(D) $\cot^{-1}|t|$

80. A curve C has the property that if the tangent drawn at any point P on C meets the coordinate axes at A and B, then P is the mid-point of AB. The curve passes through the point (1, 1). The equation of the curve is

(A) $xy = 1$

(B) $y^2 = 2x - 1$

(C) $x^2 = 2y - 1$

(D) $xy = 4x - 3$

81. For the curve $x^2y^3 = c$ (where c is a constant) the portion of the tangent between the axes is divided in the ratio

(A) 3 : 5 (B) 2 : 5

(C) 3 : 2 (D) 1 : 5

82. The number of points where the curve $y = \sin x + \frac{1}{2}\sin 2x + \frac{1}{4}\sin 4x$ has a horizontal tangent in the interval $(0, \pi)$ is

(A) 1 (B) 2

(C) 3 (D) 4

83. Equation of normal drawn to the graph of the functions defined as $f(x) = \frac{\sin x^2}{x}$, $x \neq 0$ and $f(0) = 0$ at the origin is

(A) $x + y = 0$ (B) $x - y = 0$

(C) $y = 0$ (D) $x = 0$

84. Consider the curve represented parametrically by the equation $x = t^3 - 4t^2 - 3t$ and $y = 2t^2 + 3t - 5$, where $t \in \mathbb{R}$. If H denotes the number of point on the curve where the tangent is horizontal and V the number of point where the tangent is vertical, then

(A) $H = 2$ and $V = 1$ (B) $H = 1$ and $V = 2$

(C) $H = 2$ and $V = 2$ (D) $H = 1$ and $V = 1$

85. The ordinate of all points on the curve $y = \frac{1}{2\sin^2 x + 3\cos^2 x}$ where the tangent is horizontal, is

(A) Always equal to $\frac{1}{2}$ (B) Always equal to $\frac{1}{3}$

(C) $\frac{1}{2}$ or $\frac{1}{3}$ (D) $\frac{1}{6}$

86. Equation of tangent to the curve $y = \sqrt{101 - (\sqrt{-x})^4}$ at the point where the curve intersects another curve $y = \log_{10}|x|$ is

(A) $y + 10x = 101$ (B) $y - 10x = 101$

(C) $x + 10y = 101$ (D) $x - 10y = 101$

87. Let a curve $y = f(x)$, $f(x) \geq 0 \ \forall \ x \in R$ has property that for every point P on the curve length of subnormal is equal abscissa of P. If $f(1) = 3$, then $f(4)$ is equal to

(A) $-2\sqrt{6}$ (B) $2\sqrt{6}$

(C) $3\sqrt{5}$ (D) $-3\sqrt{5}$

88. The length of the normal to the curve $x = a(\theta + \sin\theta)$, $y = a(1 - \cos\theta)$, at $\theta = \frac{\pi}{2}$ is

(A) $2a$ (B) $a\sqrt{2}$

(C) $\frac{a}{2}$ (D) $\frac{a}{\sqrt{2}}$

89. The tangent to the curve $3xy^2 - 2x^2y = 1$ at (1, 1) meets the curve again at the point

(A) $\left(-\frac{16}{5}, -\frac{1}{20}\right)$ (B) $\left(\frac{16}{5}, \frac{1}{20}\right)$

(C) $\left(\frac{1}{20}, \frac{16}{5}\right)$ (D) $\left(-\frac{1}{20}, \frac{16}{1}\right)$

90. A particle describes an ellipse whose semi-axes are 4 m and 3 m with centre at origin and a constant speed of 1 m/s. The velocity of the foot of the perpendicular from the particle on the major axis, when the particle is at a distance of 1 m from the major axis is equal to

(A) $\frac{2}{11}$ m/s (B) $\frac{11}{2}$ m/s

(C) $\sqrt{\left(\frac{2}{11}\right)}$ m/s (D) $\sqrt{\frac{7}{11}}$ m/s

91. If the curve $\frac{x^2}{a^2} + \frac{y^2}{4} = 1$ and $y^3 = 16x$ intersect at right angles, then values of a is/are

(A) $\frac{2}{\sqrt{3}}$ (B) 2

(C) $-\frac{2}{\sqrt{3}}$ (D) –2

92. The equation of normal to the curve $\left(\frac{x}{a}\right)^n + \left(\frac{y}{b}\right)^n = 2 (n \in \mathbb{N})$ at the point with abscissa equal to a can be

(A) $ax + by = a^2 - b^2$ (B) $ax + by = a^2 + b^2$

(C) $ax - by = a^2 - b^2$ (D) $bx - ay = a^2 - b^2$

93. For the curve represented parametrically by the equations, $x = 2 \ln \cot t + 1$ and $y = \tan t + \cot t$

(A) Tangent at $t = \frac{\pi}{4}$ is parallel to x-axis (B) Normal at $t = \frac{\pi}{4}$ is parallel to y-axis

(C) Tangent at $t = \frac{\pi}{4}$ is parallel to the line $y = x$ (D) Tangent and normal intersect at the point (2, 1)

94. For function $f(x) = \frac{\ln x}{x}$, which of the following statements are true?

(A) $f(x)$ has horizontal tangent at $x = e$

(B) $f(x)$ cuts the x-axis only at one point

(C) $f(x)$ is many one function

(D) $f(x)$ has one vertical tangent

95. A curve with equation of the form $y = ax^4 + bx^3 + cx + d$ has zero gradient at the point (0, 1) and also touches the x-axis at the point (–1, 0). Then the values of x for which the curve has negative gradient are

(A) $x > -1$ for $a < 0$ (B) $x < 1$

(C) $x < -1$ for $a > 0$ (D) $-1 \le x \le 1$

Paragraph for Q. Nos. 96 to 98

Let $f(x)=\frac{1}{1+x^2}$. Let m be the slope, a be the x-intercept and b be the y-intercept of a tangent to $y = f(x)$, then

96. Abscissa of the point of contact of the tangent for which m is greatest

(A) $\frac{1}{\sqrt{3}}$ (B) 1

(C) –1 (D) $-\frac{1}{\sqrt{3}}$

97. The greatest value of b is equal to

(A) $\frac{9}{8}$

(B) $\frac{3}{6}$

(C) $\frac{1}{8}$

(D) $\frac{5}{8}$

98. The abscissa of the point of contact of tangent for which m is smallest is equal to

(A) $\frac{1}{\sqrt{3}}$ (B) 1

(C) –1 (D) $-\frac{1}{\sqrt{3}}$

99. Match the following.

Column-I	**Column-II**
A. Circular plate is expanded by heat from radius 5 cm to 5.06 cm. The approximate increase in area is	(P) 4
B. Sides of a cube increasing by 1%, then the percentage increase in volume is	(Q) 0.6π
C. If the rate of decrease of $\frac{x^2}{2}-2x+5$ is twice the rate of decrease of x, then x is equal to	(R) 7
D. Rate of increase in area of equilateral triangle of side 15 cm, when each side is increasing at the rate of 0.1 cm/s is	(S) $\frac{3\sqrt{3}}{4}$

100. Match the following.

	Column-I	Column-II
A.	If $f(x)$ is a quadratic expression in x and $6\int_0^1 f(x)\,dx = k\cdot f(1)+f(0)+4f\left(\frac{1}{2}\right)$, then k is	(P) 1
B.	If the sum of the squares of the intercepts on the axes cut-off by the tangents to the curve $x^{1/3}+y^{1/3}=a^{1/3}$ $(a>0)$ at $\left(\frac{a}{8},\frac{a}{8}\right)$ is 2, then a is	(Q) 4
C.	The distance of the point on $y=x^4+3x^2+2x$ which is nearest to the line $y=2x-1$ is r, then $\sqrt{5}r$ is	(R) 3
D.	If $f(x)=-2\lvert\sin x\rvert+ae^{\lvert x\rvert}+5\lvert\tan x\rvert^3$ is differentiable at $x=0$, then a is	(S) 2

101. Let $x = a\cos t + at\sin t$ and $y = a\sin t - at\cos t$. Match the columns for $t=\frac{\pi}{4}$

	Column-I	Column-II
A.	Length of sub-tangent	(P) $\frac{a\pi}{2}$
B.	Length of normal	(Q) $\frac{a}{\sqrt{2}}\left(1-\frac{\pi}{4}\right)$
C.	Length of intercepts by tangent between coordinate axes	(R) $\frac{a\pi}{4}$
D.	Perpendicular distance of tangent from origin	(S) $a\left(1-\frac{\pi}{4}\right)$

102. Find the least positive integer value of α for which Rolle's Theorem is applicable to function

$$f(x)=\begin{cases}x^{(\alpha^2-4\alpha+3)}\cdot\log x, & \text{if } x>0\\ 0, & \text{if } x=0\end{cases} \text{ in } [0,\ 1]$$

103. From a point P on the curve $a^4x + 2b^2y^2 = 0$, tangent PQ and PR are drawn to hyperbola $\frac{x^2}{b^2}-\frac{y^2}{a^2}=1$. If QR touches a fixed parabola, then find the length of latus rectum of that parabola.

104. Let $f(x)$ be a differentiable function such that $f'(x)=\frac{1}{\log_3[\log_{1/4}(\cos x+a)]}$. If $f(x)$ is increasing for all values of x, then

(A) $a\in(5,\infty)$ (B) $a\in\left(1,\frac{5}{4}\right)$

(C) $a\in\left(\frac{5}{4},5\right)$ (D) $a\in\phi$

105. Let $f:[2,7]\to[\mathbb{R}]$ be a continuous and differentiable function. Then the value of $(f(7)-f(2))\dfrac{(f(7))^2+(f(2))^2+f(2)\cdot f(7)}{3}$, is (where $c\in(2,7)$)

(A) $3f^2(c)\cdot f'(c)$ (B) $5f^2(c)\cdot f(c)$

(C) $5f^2(c)\cdot f'(c)$ (D) $f^2(c)\cdot f'(c)$

106. Let $f(x)=ax^5+bx^4+cx^3+dx^2+ex$, where $a, b, c, d, e\in\mathbb{R}$ and $f(x)=0$ has a positive root α, then

(A) $f'(x)=0$ has root α_1 such that $0<\alpha_1<\alpha$ (B) $f''(x)=0$ has at least one real root

(C) $f'(x)=0$ has at least two real roots (D) All of these

107. If $f(x)$ is a polynomial of degree 5 with real coefficients such that $f(|x|)=0$ has 8 real roots, then $f(x)=0$ has

(A) 4 real roots only (B) 5 real roots (one –ve other +ve)

(C) 3 real roots (D) Nothing can be said

108. Consider $f(x)=\int_0^x\left(t+\dfrac{1}{t}\right)dt$ and $g(x)=f'(x)$ for $x\in\left[\dfrac{1}{2},3\right]$. If P is a point on the curve $y=g(x)$ such that the tangent to this curve at P is parallel to a chord joining the points $\left(\dfrac{1}{2},g\left(\dfrac{1}{2}\right)\right)$ and $(3, g(3))$ of the curve, then the coordinates of the point P

(A) Can't be found out (B) $\left(\dfrac{7}{4},\dfrac{65}{28}\right)$

(C) (1, 2) (D) $\left(\sqrt{\dfrac{3}{2}},\dfrac{5}{\sqrt{6}}\right)$

109. A function $g(\theta)=\int_0^{\sin^2\theta}f(x)dx+\int_0^{\cos^2\theta}f(x)dx$ is defined in the interval $\left(-\dfrac{\pi}{2},\dfrac{\pi}{2}\right)$, where $f(x)$ is an increasing function, then $g(\theta)$ is increasing in the interval

(A) $\left(-\dfrac{\pi}{2},0\right)$ (B) $\left(-\dfrac{\pi}{2},-\dfrac{\pi}{4}\right)$

(C) $\left(0,\dfrac{\pi}{4}\right)$ (D) $\left(-\dfrac{\pi}{4},0\right)$

110. From a fixed point P on the circumference of a circle of radius a, the perpendicular PR is drawn to the tangent at Q(a variable point). The maximum area of ΔPQR is ka^2. Then k is

(A) $\dfrac{3\sqrt{3}}{4}$ (B) $\dfrac{3\sqrt{3}}{8}$

(C) $\dfrac{2\sqrt{3}}{5}$ (D) $\dfrac{2\sqrt{2}}{3}$

111. The lateral edge of a regular rectangular pyramid is a cm long. The lateral edge makes an angle α with the plane of the base. The value of α for which the volume of the pyramid is greatest is

(A) $\dfrac{\pi}{4}$ (B) $\sin^{-1}\sqrt{\dfrac{2}{3}}$

(C) $\cot^{-1}\sqrt{2}$ (D) $\dfrac{\pi}{3}$

112. A curve with equation of the form $y = ax^4 + bx^3 + cx + d$ has zero gradient at the point (0, 1) and also touches the x-axis at the point (–1, 0), then the values of x for which the curve has a negative gradient are

(A) $x > -1$ (B) $x < 1$

(C) $x < -1$ (D) $-1 \le x \le 1$

113. P and Q are two points on a circle of centre C and radius α, the angle PCQ being 2θ, then the radius of the circle inscribed in the triangle CPQ is maximum when

(A) $\sin\theta = \frac{\sqrt{3}-1}{2\sqrt{2}}$ (B) $\sin\theta = \frac{\sqrt{5}-1}{2}$

(C) $\sin\theta = \frac{\sqrt{5}+1}{2}$ (D) $\sin\theta = \frac{\sqrt{5}-1}{4}$

114. Let $f(x) = \int_1^x \left(t\ln(t) - \frac{\ln t}{t} \right) dt;\ (x > 1)$, then

(A) $f(x)$ has one point of maxima and no point of minima

(B) $f'(x)$ has two distinct roots

(C) $f(x)$ has one point of minima and no point of maxima

(D) $f(x)$ is monotonic

115. The lateral edge of a regular hexagonal pyramid is 1 cm. If the volume is maximum, then its height must be equal to

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$

(C) $\frac{1}{\sqrt{3}}$ (D) 1

116. Two points $A(1, 4)$ and $B(3, 0)$ are given on the ellipse $2x^2 + y^2 = 18$. The coordinates of a point on the ellipse such that the area of the triangle ABC is greatest is equal to

(A) $(\sqrt{6}, \sqrt{6})$

(B) $(-\sqrt{6}, \sqrt{6})$

(C) $(\sqrt{6}, -\sqrt{6})$

(D) $(-\sqrt{6}, -\sqrt{6})$

117. The minimum value of the function $f(x) = \frac{x^p}{p} + \frac{x^{-q}}{q}$, where $\frac{1}{p} + \frac{1}{q} = 1,\ p > 1$ is equal to

(A) 1 (B) 0

(C) 2 (D) 5

118. If $f : \mathbb{R} \to \mathbb{R}$, $f(x)$ is a differentiable bijective function, then which of the following may be true?

(A) $(f(x) - x)\, f''(x) < 0\ \forall\ x \in \mathbb{R}$

(B) $(f(x) - x)\, f''(x) > 0\ \forall\ x \in \mathbb{R}$

(C) If $(f(x) - x)\, f''(x) > 0$, then $f(x) = f^{-1}(x)$ has no solution

(D) If $(f(x) - x)\, f''(x) > 0$, then $f(x) = f^{-1}(x)$ has at least one real solution

119. Which of the following statements is true about the monotonocity of $Q(x)$, where $Q(x) = 2f\left(\frac{x^2}{2}\right) + f(6 - x^2)\ \forall\ x \in R$? If it is given that $f''(x) > 0\ \forall\ x \in R$

(A) $Q(x)$ is increasing in $x \in (-2, 0) \cup (2, \infty)$

(B) $Q(x)$ is decreasing in $x \in (-\infty, -2) \cup (0, 2)$

(C) $Q(x)$ is increasing in $x \in (-\infty, -2) \cup (-2, 0)$

(D) $Q(x)$ is decreasing in $x \in (-2, 0) \cup (2, \infty)$

120. The function $f(x) = \int_0^x \sqrt{1 - t^4}\, dt$ is such that

(A) It is defined on the interval [–1, 1]

(B) It is an increasing function

(C) It is an odd function

(D) It is an even function

121. For the function $f(x) = x^4\ (12 \ln x - 7)$

(A) The point (1, –7) is the point of inflection

(B) $x = e^{1/3}$ is the point of minima

(C) The graph is concave downwards in (0, 1)

(D) The graph is concave upwards in (1, ∞)

122. Let $f(x) = (x - 1)^4 \cdot (x - 2)^n$, $n \in \mathbb{N}$. Then $f(x)$ has

(A) A maximum at $x = 1$, if n is odd

(B) A maximum at $x = 1$, if n is even

(C) A minimum at $x = 2$, if n is even

(D) A maximum at $x = 2$, if n is odd

Paragraph for Q. Nos. 123 and 124

$f : \mathbb{R} \to \mathbb{R}$, $f(x)$ is a differentiable function such that all its successive derivatives exist, $f'(x)$ can be zero at discrete points only and $f(x) \cdot f''(x) \le 0\ \forall\ x \in \mathbb{R}$.

123. If α and β are two consecutive roots of $f(x) = 0$, then

(A) $f''(\gamma) = 0$, $\gamma \in (\alpha, \beta)$ for some γ

(B) $f'''(\gamma) = 0$, $\gamma \in (\alpha, \beta)$ for some γ

(C) $f''''(\gamma) = 0$, $\gamma \in (\alpha, \beta)$ for some γ

(D) $f''''(\gamma) = 0$, $\gamma \in (\alpha, \beta)$ for some γ

124. If $f'(x) \neq 0$, then maximum number of real roots of $f''(x) = 0$ is/are

(A) No real root

(B) One

(C) Two

(D) Three

Paragraph for Q. Nos. 125 to 127

If $f(x)$ is a differentiable function whether it is continuous and $f'(c_1) = f'(c_2) = 0$, $f''(c_1) \cdot f''(c_2) < 0$, $f(c_1) = 5$, $f(c_2) = 0$ and $(c_1 < c_2)$.

125. If $f(x)$ is continuous in $[c_1, c_2]$ and $f''(c_1) - f''(c_2) > 0$, then minimum number of roots of $f'(x) = 0$ in $[c_1 - 1, c_2 + 1]$ is

(A) 2

(B) 3

(C) 4

(D) 5

126. If $f(x)$ is continuous in $[c_1, c_2]$ and $f''(c_1) - f''(c_2) < 0$, then minimum number of roots of $f'(x) = 0$ in $[c_1 - 1, c_2 + 1]$ is

(A) 1

(B) 2

(C) 3

(D) 4

127. If $f(x)$ is continuous in $[c_1, c_2]$ and $f''(c_1) - f''(c_2) > 0$, then minimum number of roots of $f(x) = 0$ in $[c_1 - 1, c_2 + 1]$ is

(A) 2

(B) 3

(C) 4

(D) 5

128. Match the columns.

	Column-I	Column-II
A.	The number of the distinct real roots of the equation $(x + 1)^5 = 2(x^5 + 1)$ is	(P) 16
B.	The absolute maximum value of the function $f(x) = \frac{(x+1)^4}{x^4 - x^3 + x^2 - x + 1}$ is	(Q) 3
C.	Let $f(x) = ab\sin x + \sqrt{1-a^2}\cos x + c$, where $\lvert a\rvert, \lvert b\rvert < 1$, then difference of maximum and minimum value of $f(x)$ is	(R) 1
D.	If $u = \sqrt{4\cos^2\theta + \sin^2\theta} + \sqrt{4\sin^2\theta + \cos^2\theta}$, then the difference between maximum and minimum value of u^2 is	(S) 2

129. Match the columns.

	Column-I	Column-II
A.	A function f is differentiable in [0, 5] such that $f(0) = 4$ and $f(5) = -1$. If $g(x) = \frac{f(x)}{x+1}$, then there exists some $c \in (0, 5)$ such that $g'(c)$ is equal to	(P) 8
B.	Let $f(x)$ and $g(x)$ be differentiable for $0 \le x \le 1$, $f(0) = 2$, $g(0) = 0$, $f(1) = 6$. Let there exists a real number $c \in (0, 1)$ such that $f'(c) = 2g'(c)$, then $g(1)$ is equal to	(Q) $\frac{-5}{6}$
C.	Let f be differentiable $\forall\ x$, if $f(1) = -2$, $f'(x) \ge 2\ \forall\ x \in [1, 6]$, then $f(6)$ has least possible value	(R) 15
D.	Let Lagrange's mean value theorem is satisfied for $f(x) = \sqrt{25 - x^2}$ and $c \in (1, 5)$. Then the value of c^2 is	(S) 2

130. A circular sheet has radius $\sqrt{6}$ units. A sector is cut and the remaining part of sheet is used to form a cone by joining the edges. Let V_0 be the maximum volume of which the cone can be formed, then evaluate $\tan^2\left(\frac{V_0}{8\sqrt{2}}\right)$.

131. Let $f(x)$ be a function such that $f(x + y) = f(x) + f(y) + 2xy - 1\ \forall\ x, \in \mathbb{R}$ and $f'(0) = 2$. Further let $g(x) = \int_0^x f(t)\,dt$, then find the maximum value of $g(x)$ in [1, 3].

132. If $f(x)$ is a twice differentiable function in $\mathbb{R}$ such that $f(a) = 0$, $f(b) = -2$, $f(c) = 3$, $f(d) = -1$, $f(e) = 0$; $a < b < c$, then find the minimum number of roots of equation $g(x) = 0$, where $g(x) = (f'(x))^2 + f(x) \cdot f''(x)$.

133. If $a, b > 0$ such that $a^2 + b = 2^{5/3} \cdot 3^{1/3}$, then find the maximum value of term independent of x in the expansion of $\left(\frac{a}{2}x^{1/6} + \frac{b}{3}x^{-1/3}\right)^9$.

134. For x to be real, find the maximum value of $f(x) = 6(\cos x - x)\left(x + \sqrt{x^2 + \sin^2 x}\right)$.

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